Venus Photochemistry

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by

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Abstract

Something in the clouds of Venus absorbs near-ultraviolet sunlight and is responsible for about half of the solar heating at the cloud top, yet nearly a century after its markings were first photographed, the substance is still unknown. This project asks not which molecule has the right spectrum but whether any proposed carrier can be present in the right amount, at the right altitude and with the right spectral shape, all at once, inside a model that is free to move it around.

The work uses VULCAN-Venus, a one-dimensional model of the photochemistry of the Venus atmosphere. Phase 1 found that the model's absorber term was present but switched off, switched it on under a single amplitude, and ran fourteen experiments on the model and on published laboratory data. Every number is recomputed from the saved model output by one script into a registry of 10830 values, and a second script checks every printed number precise enough to test against it.

The absorber's effect on the radiation is proportional to its amount and is concentrated in one band of altitude, 69 to 93 km, which holds 91.370% of the response. Where an absorber sits matters more than its spectrum: moved up, the same absorber changes the radiation 7.34 times more than moved down, against a factor of 4.95 between the spectra tested at a fixed height. When the model's own chemistry is allowed to place ClS_2, the gas closest to the needed amount, its centre falls 51.521 km lower and it becomes radiatively the same as no absorber, to 0.001487%. A screen of all 23 species the model can represent finds none that passes amount, placement and spectrum together. From published data, the proposed iron and sulfur minerals cannot absorb enough at 375 nm even as pure solids, and a published rejection of ferric chloride, based on a measurement in water, does not hold if it absorbs in the droplets as it does in hydrochloric acid, where the amount needed falls from 71.0% of the droplet mass to between 2.82% and 17.05%. Changes in abundance smaller than 2.5278% are shown to be numerical, not physical, and are reported only as upper bounds.

Phase 2 proposes to test these verdicts under the much stronger vertical mixing now estimated for the Venus clouds, to ask whether upwelling can carry a candidate into the response band, and to place ferric chloride by the same altitude test.

Keywords: Venus, unknown ultraviolet absorber, photochemistry, VULCAN, radiative transfer, ferric chloride, reproducibility

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List of Terms

Absorber grid. 300 to 698 nm, the wavelengths over which the absorber term acts (Chapter 4).

Absorption efficiency, Q_{abs}. The fraction of the light falling on a particle's cross-sectional area that the particle absorbs; the absorber's spectrum is a table of it (Chapter 3).

Actinic flux. The number of photons per square centimetre per second per nanometre reaching a point from all directions; it sets every photolysis rate (Chapter 3).

Amplitude, A_{UUV}. The single number that scales the absorber term: 0 for the control, 1 for the supplied absorber, and 0.5 and 2 in Experiment 3 (Chapters 3 and 5).

Baseline and reference. The reference is the output supplied with the model, kept untouched; the baseline is the same model run afresh on the project computer (Chapter 3).

Bounded mean. A mean over a field that leaves out the elements where both values are zero (Chapter 4).

Cloud top. The height at which the cloud's optical depth, counted downward from space, reaches one (Chapter 1).

Column and column centre. A quantity per cubic centimetre added over altitude with each layer thickness in centimetres, giving a quantity per square centimetre; the column centre, or centroid, is the mean altitude weighted by that quantity (Chapters 3 and 4).

Control and treatment. Two runs that differ only in the absorber: switched off in the control, switched on in the treatment (Chapter 3).

Cross section. The effective area one molecule presents to light, in square centimetres (Chapter 3).

Decadic and Napierian. Absorbance measured with base-10 and with natural logarithms; the two differ by a factor of $\ln 10$ (Chapters 2 and 6).

Eddy diffusion coefficient, K_{zz}. The strength of vertical mixing in the model (Chapters 2 and 3).

Empirical absorber. The absorber supplied with the model, a spectrum and a height profile fitted to the observed brightness of Venus (Chapters 1 and 3).

Enhancement band. 320 to 400 nm, the near-ultraviolet feature the absorber has to supply (Chapter 4).

Field change, ΔF. The size of the change in the whole actinic flux field, as a percentage of the size of the reference field (Chapter 4).

Isomers. Molecules made of the same atoms arranged differently, such as ClSSCl and $SSCl_2$ (Chapters 2 and 6).

Loading. The mass of a candidate needed per litre of droplet to meet the absorption requirement (Chapter 6).

Mass absorption coefficient, ε. How strongly a material absorbs per gram, in square centimetres per gram; dividing the required absorption by it gives the loading (Chapter 6).

Mixing ratio. The number density of a species divided by that of the air, quoted in parts per million (ppmv) or per billion (ppb) (Chapters 2 and 3).

Mode 1 droplets. Cloud droplets smaller than one micrometre across, in which the absorber is thought to sit (Chapter 1).

Optical depth. A measure of how much light a layer removes; light crossing an optical depth of one keeps a fraction e^{-1} of its intensity (Chapters 1 and 3).

Photolysis. The breaking of a molecule by a photon. Its rate coefficient J is the probability per second that one molecule is broken; the rate Jn also depends on how much of the molecule is present (Chapters 3 and 4).

Radiation floor. 0.0017%: how far the radiation fields of two runs that should agree differ, as a share of the change Experiment 9 makes, not of the whole field (Chapters 4 and 6).

Reflectivity, I/F. The observed radiance divided by $\pi F_{\odot}$, where $F_{\odot}$ is the solar flux arriving at Venus (Chapter 1).

Registry and key. The file `verify/registry.json` that holds every number in this report, and the dotted name of each value in it (Chapter 8).

Relative change. A difference divided by the larger of the two values, not by the control (Convention 1, Chapter 4).

Reproducibility floor, or chemistry floor. 2.5278%: in-band changes in abundance smaller than this are numerical, not physical, and are reported only as upper bounds

(Chapter 4).

Requirement constant, τ_{req}. The empirical absorber's optical depth through the whole atmosphere at 320 nm, the amount of absorption any candidate must supply (Chapter 3).

Response band and centroid. The narrowest range of altitude, 69 to 93 km, that holds at least 90 per cent of the radiative response, and the weighted mean altitude of that response (Chapter 4).

Screening window. 300 to 620 nm, the range searched for each candidate's strongest absorption (Chapter 4).

Shortfall factor. How many times too little of a species the model holds to supply the required optical depth (Chapter 6).

Single scattering albedo. The fraction of light meeting a particle that is scattered rather than absorbed (Chapter 2).

Spectral correlation. How closely a candidate's spectrum follows the empirical absorber's, as a correlation coefficient between -1 and 1 (Chapter 6).

Steady state. The state in which production, destruction and transport of every species balance, tested by `longdy` and `longdydt` (Chapters 1 and 3).

Super-rotation. The flow of the atmosphere at the cloud top round the planet far faster than the planet itself turns (Chapter 1).

Weight per cent, wt%. The share of a mixture's mass that one component makes up (Chapter 2).

Chapter 1

Introduction

This report describes Phase 1 of an M.Tech. project on the atmosphere of Venus. It starts from the first reading of the literature and ends with the proposal for Phase 2. The subject is a single long-standing problem: something in the clouds of Venus absorbs near-ultraviolet sunlight, and nobody knows what it is. This chapter explains what is known about that problem, what has been tried, and how this project approaches it differently.

1.1 Venus and its clouds

Venus has a thick atmosphere. The pressure at the surface reaches 92 bar and the surface temperature is about 750 K [5]. The planet is completely covered by a dense cloud deck that lies between 47 and about 70 km above the surface and is more than 20 km thick [5]. The clouds are made of small droplets of concentrated sulfuric acid, H_2SO_4 [5].

Measurements made by a descent probe divide the cloud deck into three layers by the mass of cloud material they hold: the lower cloud, from about 48 to 50 km; the middle cloud, from about 50 to 56 km; and the upper cloud, from about 56 to 70 km [5]. The droplets are also grouped by size into *modes*. Mode 1 droplets are smaller than one micrometre across and are the most numerous throughout the cloud deck; mode 2 droplets are about two micrometres across [5]. Mode 1 matters for this report because the unknown absorber is now thought to be mixed into mode 1 droplets [5, 13].

The *cloud top* is defined as the height at which the optical depth of the cloud, counted downward from space, reaches one [5]. Optical depth is a measure of how much light a layer removes: light that crosses an optical depth of one keeps a fraction e^{-1}, about a third, of its original intensity. Depending on the wavelength used and the place observed, the cloud top lies between 65 and 72 km [5].

The solid planet turns very slowly. One rotation relative to the stars takes 243

Earth days, and one solar day, from noon to noon, takes 117 Earth days [1]. The atmosphere at the cloud top moves far faster. It travels westward round the planet in about four days, with a wind speed of about 100 metres per second [1]; in the cloud layer as a whole the circuit takes 4 to 6 days [12]. This fast circulation is called *super-rotation*. It was first revealed by ultraviolet observations of the cloud markings (Ross, 1928, and Boyer and Camichel, 1961, as cited in [13]), which is the first sign that the ultraviolet appearance of Venus and its atmospheric motion are linked.

Figure 1.1 summarises this structure. It also shows two things that come from the present project and are explained later: the grey band, which marks the altitudes where the model's radiation responds to the absorber (Chapter 5), and the altitudes where the model's own chemistry places a set of candidate molecules (Chapter 6).

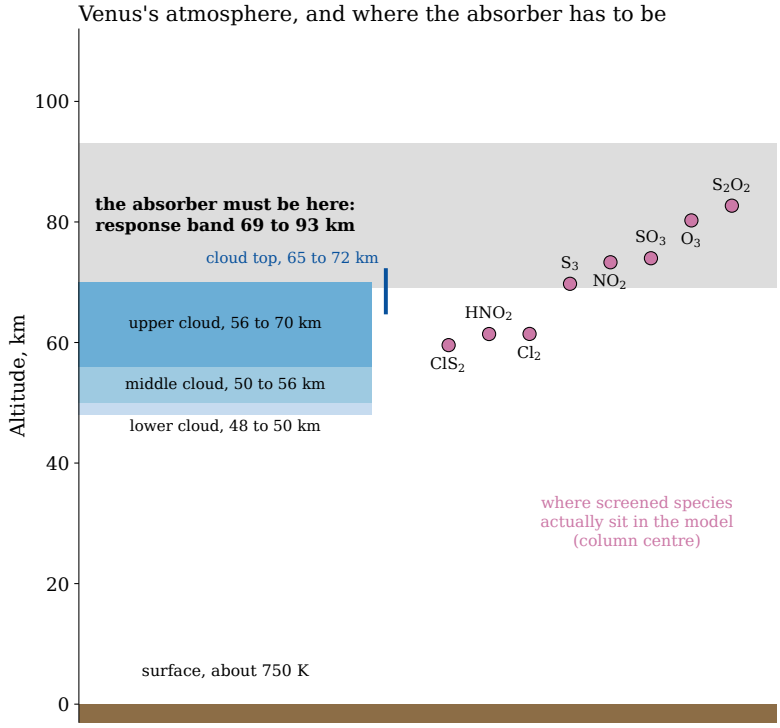


Figure 1.1: The Venus atmosphere, and where the absorber has to be. Left: the three cloud layers and the range of cloud top heights, from [5]; the surface temperature, from the same source. Grey band: the response band, 69 to 93 km, found in Experiment 6 (Chapter 5). Purple circles: the column centre of eight candidate molecules as placed by the model's own chemistry (Experiment 12). Sources: `figures/literature_values.json`; registry keys `e6.interval.90.lo`, `e6.interval.90.hi` and `e12.<species>.centroid`.

1.2 The unknown ultraviolet absorber

In visible light Venus is very bright: its clouds scatter about 80 per cent of the incoming sunlight back to space [5]. Towards the ultraviolet the planet becomes darker, and the darkening has two parts. Below 320 nm the shape of the absorption

matches sulfur dioxide, SO_2, which is therefore taken to be responsible for it [5]. But SO_2 absorbs only weakly at longer wavelengths, and its effect is negligible beyond 320 nm [13]. Between 320 and 400 nm there is a second, strong absorption that no known species has been shown to explain [5]. In the spectrum of the planet it appears as a broad band reaching from the near ultraviolet into the blue [13]. Because its carrier is not known, it is called the *unknown ultraviolet absorber* [5]. In this report it is called simply *the absorber*.

Figure 1.2 shows the problem directly. The upper panel is one reflectivity spectrum of Venus, number 151, measured by the MESSENGER spacecraft during its second Venus flyby on 5 June 2007 [13]. Reflectivity, written I/F, is the observed radiance divided by $\pi F_\odot$, where $F_\odot$ is the solar flux arriving at Venus [13]: a perfectly white, evenly scattering surface lit from directly overhead would have $I/F = 1$. The lower panel is the absorption cross section of SO_2 used in the model, which measures how strongly one SO_2 molecule absorbs at each wavelength. Venus is dark between 320 and 400 nm, which is exactly where SO_2 has almost stopped absorbing. From the same MESSENGER data, Pérez-Hoyos et al. found the absorption to be centred near 340 nm with a full width at half maximum of about 140 nm [13].

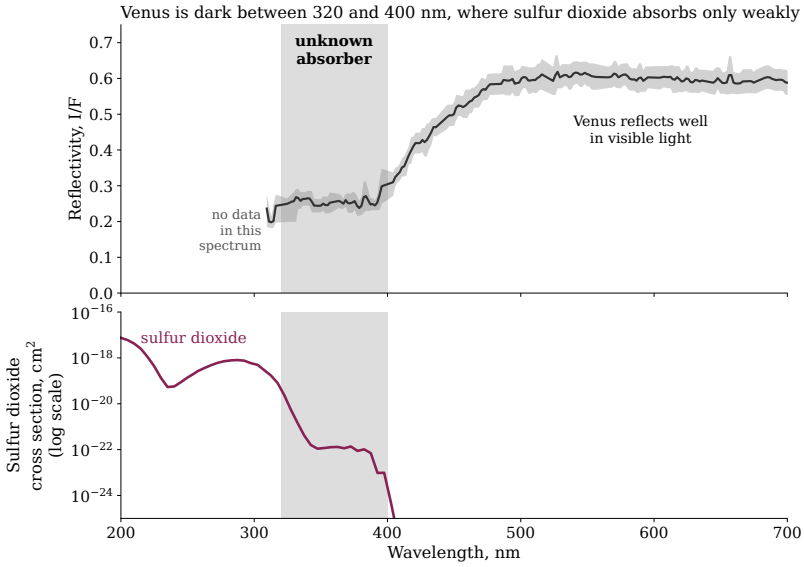


Figure 1.2: Venus is dark between 320 and 400 nm, where sulfur dioxide absorbs only weakly. Top: observed reflectivity I/F of MESSENGER spectrum 151, digitised from Figure 8 of Pérez-Hoyos et al. [13]; the shaded envelope is the drawn observation with its plotted error range, and the line its middle. Bottom: the SO_2 absorption cross section used in the model, on a logarithmic scale. Sources: `figures/data/perez-hoyos-2018-fig8-IF.csv` (with its provenance file) and `thermo/photo_cross/SO2/SO2_cross.csv`.

The absorber matters for three reasons.

It heats the cloud top. Observations and models agree that the absorber is respon-

sible for about half of the solar heating at the cloud top [5]. Solar heating in the upper clouds has in turn been proposed as one of the engines of the super-rotation, through the thermal tides it excites [13].

It changes over time, and the circulation changes with it. The brightness of Venus at 365 nm varied strongly between 2006 and 2017, which implies that the amount of absorber varied by a factor of two; the resulting change in solar heating has been proposed to alter the strength of the Hadley circulation, the large overturning flow between the equator and the poles (Lee et al., 2019, as reviewed in [5]).

Its pattern traces vertical motion. The dark and bright markings seen in ultraviolet images are not random. At low latitudes the clouds are dark because convection brings the absorber up from below; at middle latitudes a cold layer with a stable temperature profile suppresses vertical mixing and cuts off that supply, so the clouds there are bright and uniform [16]. In other words, where the absorber appears depends on how the atmosphere moves it up and down. This observation is the starting point of the present project.

What is known about *where* the absorber sits is limited. Measurements of sunlight made by descent probes found that ultraviolet absorption happens mainly in the upper cloud, above about 57 km, and is negligible in the middle and lower clouds (Tomasko et al., 1980, and Ekonomov et al., 1984, as reviewed in [5]). The supply of absorber from below appears to be continuous at low latitudes [5]. Nearly a century after the ultraviolet contrasts were first photographed from Earth, the substance that causes them is still unknown [12].

1.3 The candidates proposed so far

Many substances have been put forward as the absorber. Table 1.1 lists the main ones, as summarised in a recent review [5] and in the MESSENGER study [13]. The table is a summary for orientation; Chapter 2 reviews the key papers one by one.

Most of these proposals share one form of argument. A substance is measured in the laboratory, or its spectrum is calculated, and the spectrum is compared with the observed absorption. If the two look alike, the substance is put forward as a candidate. This argument establishes one thing: that the substance absorbs at the right wavelengths. It does not establish that enough of the substance can exist on Venus, and it does not establish that the substance sits at the altitude where the absorption happens.

Those two further questions have been asked for a few candidates, one at a time.

Table 1.1: The main candidates for the unknown absorber, what supports each, and what counts against it. Proposers are given as cited in the sources in the last column.

Candidate	In its favour	Against it	Source
Sulfur dioxide, SO_2	Matches the absorption below 320 nm.	Negligible beyond 320 nm, so it cannot supply the band between 320 and 400 nm.	[13, 5]
Polysulfur, S_x (Hapke and Nelson, 1975)	One of the first candidates considered for its spectral properties.	No single form matches the observed spectrum; it does not dissolve in sulfuric acid and would spoil an optical effect, the glory, that the droplets show.	[5]
Ferric chloride, $FeCl_3$, dissolved in the droplets (Zasova et al., 1981)	Spectral behaviour close to the absorber; dissolves in sulfuric acid.	Its absorption band is too narrow and too weak between 400 and 500 nm.	[5]
Disulfur monoxide, S_2O (Lo et al., 2003)	Good match between 400 and 500 nm.	Its absorption weakens quickly between 320 and 340 nm, where the absorber is still strong.	[5]
Disulfur dioxide, OSSO or S_2O_2	Its two forms, cis and trans, in comparable amounts, together match the spectrum between 330 and 420 nm, a result of Frandsen et al. reviewed in [5].	A model of the clouds gives a mixing ratio at the cloud top of no more than one part in 10^{14}, far too little [4, 5]; an independent study also finds it falls short [6].	[5]
Iron and sulfur minerals: rhomboclase and acid ferric sulfate	A mixture absorbs between 300 and 500 nm, consistent with the observations [9].	Made under particular laboratory conditions; whether they exist in the clouds, and how much, is unknown.	[5]
Chlorine and sulfur compounds, such as SCl_2	Shown in the review among the candidates whose spectra come close to the absorber's.	The review gives no abundance for them in the clouds.	[5]
Organic "red oil" (Spacek and Benner, 2021)	A large organic molecule that dissolves in acid.	Listed in the review as a speculative hypothesis.	[5]
A mixture of several absorbers	Would explain the width of the band and its lack of fine structure.	Listed in the review as a possibility; not yet tested.	[5]

For OSSO, a photochemical model showed that it is destroyed almost as soon as it forms, so its abundance at the cloud top is far too small [4, 5], and an independent study reached the same conclusion [6]. For polysulfur, model calculations placed a large amount of it below the cloud base, and the same author later concluded that it cannot explain the absorption in the upper clouds [5]. The review itself concludes that a successful candidate must explain the observed spectrum, must have the right distribution of abundance in the cloud layer, and must have a stable source that keeps its abundance from being removed by chemistry and mixing [5]. What was missing at the start of this project was a way to ask these questions together, for every candidate, inside one model.

1.4 The question this project asks

The question of this project is therefore not *which molecule has the right spectrum*. It is this: **can any proposed carrier be present in the right amount, at the right altitude, and with the right spectral shape, all at once, inside a model that is free to move it around?**

Answering it needs a *photochemistry and transport model*. Such a model divides the atmosphere into horizontal layers. In each layer it tracks how fast every chemical species is made and destroyed by reactions, including reactions driven by sunlight (*photolysis*), and how fast each species is carried up and down between layers by mixing. It then runs until production, destruction and transport balance, which is called the *steady state*. The important property is that the model, not the modeller, decides where each species ends up.

This project uses VULCAN-Venus, an open-access model of this kind published by Dai, Shao and Sheng [4]. It covers the atmosphere from the surface to 112 km and follows sunlight from 100 to 700 nm [4]; the copy used here does so in 1670 wavelength bins from 96 to 698 nm (Chapter 3). The published description counts 59 species, 362 neutral reactions and 45 photolysis reactions [4]. The copy used here lists 62 entries and 838 numbered reaction slots, because its lists also hold bookkeeping items: condensed phases, two inert tracers that mark the mode 1 and mode 2 cloud particles, and a reverse slot for every reaction (Chapter 3). Chapter 3 describes it in detail. In the published model the absorber is not a chemical species. It is a fixed, empirical opacity: its amount, its height profile and its spectrum are all prescribed in advance, chosen so that the model reproduces the observed brightness of Venus. This fixed recipe is called the *empirical absorber* in this report. It is useful because it shows what the real absorber must achieve. Any candidate that is to

replace it must do the same job.

That job can be broken into three tests:

1. **Enough material.** The candidate must supply as much optical depth as the empirical absorber, wavelength by wavelength. For scale, the empirical absorber's optical depth through the whole atmosphere at 320 nm is 1.094968472081 (Chapter 3).

2. **The right altitude.** The candidate must sit where the model's radiation responds to the absorber. Experiment 6 found that this happens in a band between 69 and 93 km, called the *response band* (Chapter 5).

3. **The right spectrum.** The candidate's absorption must be concentrated between 320 and 400 nm, and its shape must follow the empirical absorber's.

Figure 1.3 shows, in advance, what happens when all three tests are applied to every species the model can represent (Experiment 12, Chapter 6). Each pass mark is one that the empirical absorber itself meets. Of 23 species that could be tested, 16 sit at the right altitude, only 2 of those also have the right spectrum, and none has enough material. The two that come closest, OSSO and HOCl, fall short at 360 nm by factors of 2.5×10^5 and 1.0×10^7.

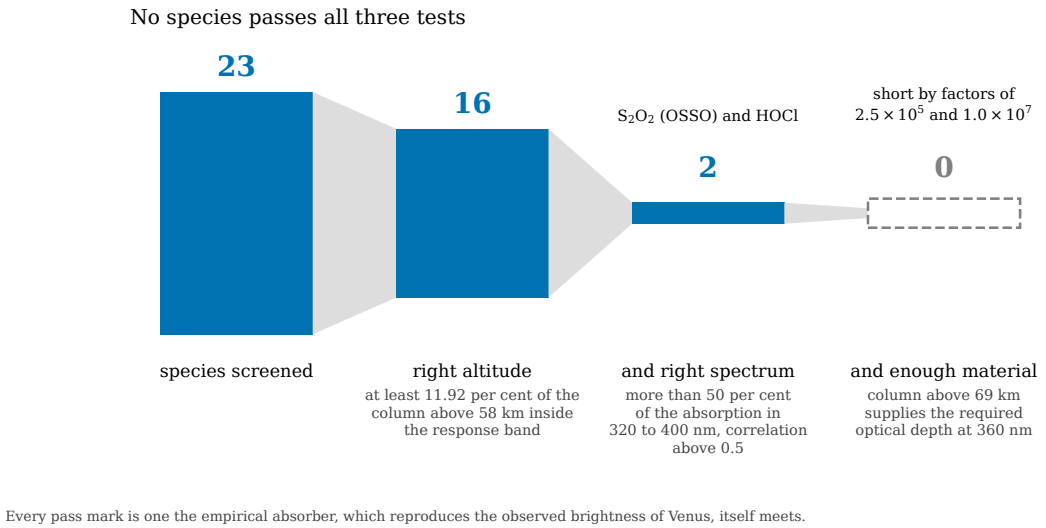


Figure 1.3: No species passes all three tests (Experiment 12). The 23 species the model can test pass through the three tests in turn; each pass mark is one the empirical absorber itself meets. The altitude test asks for at least the empirical absorber's own share of its column above 58 km inside the response band. Sources: the registry keys that begin `e12.gate`, which hold the counts and the pass marks, and `e12.S2O2.short.360` and `e12.HOCl.short.360`, which hold the two shortfalls. Chapter 8 explains how to read a registry key.

1.5 Objective and Research Gap

The objective of Phase 1 was to test, inside a photochemistry and transport model, whether the proposed carriers of the unknown absorber can meet all three requirements at once, and to find out which of the three requirements decides the outcome. It was reached through five steps, each of which became a group of experiments.

Make the absorber switchable. The model had to be able to run with the absorber on and with it off, differing in nothing else, so that any change could be blamed on the absorber alone (Chapter 3).

Find where the absorber acts. The response had to be located in amount, wavelength, chemical reactions and altitude (Experiments 1 to 6, Chapter 5).

Find what decides the size of the response. Moving the absorber up or down was compared with changing its spectrum, and the model's chemistry was then allowed to place a candidate by itself (Experiments 7 to 10, Chapters 5 and 6).

Test real candidates. Published spectra and mineral data were tested against the three requirements (Experiments 11 to 14, Chapter 6).

Verify every number. Every number in this report is recomputed from the saved model output and checked automatically (Chapter 8).

The research gap. The studies reviewed for this project test candidates mainly on spectral grounds, and the photochemical models among them treat the absorber either as a fixed opacity [4] or examine one candidate at a time [5, 6]. None of them, as far as this review found, applies the amount, the altitude and the spectrum requirements together to every species a model can represent, and none measures how strongly the radiative effect of the absorber depends on its altitude. Phase 1 fills that gap with the following main results:

- Altitude is a first-order control. Moving the absorber's profile up by 17.8439 km changes the radiation field 7.34 times more than moving it down by 18.0000 km (Experiment 7).
- The model's chemistry, not the modeller, decides where a candidate sits, and when it is allowed to decide, the candidate falls out of the response band (Experiment 9).
- No species the model can represent passes all three tests (Experiment 12), and the candidate families fail for different reasons.

- A published rejection of ferric chloride rests on an absorbance measured in water. Measured in hydrochloric acid, ferric chloride absorbs 4 to 25 times more strongly, and if it absorbs so in the droplets it becomes a plausible minor component (Experiment 14, done after Phase 1 closed). This does not identify the absorber.

These are negative results for the candidates tested, and they are reported as such.

1.6 Organisation of this report

Chapter 2 reviews the papers this project builds on. Chapter 3 describes the model and the groundwork that made controlled experiments possible. Chapter 4 states the conventions used for every number. Chapters 5 and 6 describe the fourteen experiments in the order they were done. Chapter 7 states what Phase 1 establishes and where its limits lie. Chapter 8 describes how every number was verified and how a reader can trace any number back to the code. Chapter 9 proposes Phase 2, and Chapter 10 concludes.

Chapter 2

Literature Review

This chapter reviews the twelve papers on which Phase 1 rests: the model used (Section 2.1); the absorber's observed spectrum and height and the mixing in the clouds, which decide whether a candidate can sit where the absorber is (Sections 2.2 to 2.5); and the candidates, with the one paper that states how strongly the droplets must absorb (Sections 2.6 to 2.10). Section 2.11 draws the threads together. For each paper, the first part says what the authors did and how, and the second what they found, followed by a paragraph marked as this project's own reading of what remains unsettled.

Two terms recur. A *mixing ratio* is the fraction of the molecules in a volume of air that belong to one species, in parts per million (ppmv) or per billion (ppb). The *eddy diffusion coefficient* K_{zz} is the single number per height with which a one-dimensional model represents all the vertical stirring it cannot resolve: the larger K_{zz}, the faster a species is mixed up and down. It is quoted in $\mathrm{cm^2\,s^{-1}}$ or $\mathrm{m^2\,s^{-1}}$; one $\mathrm{m^2\,s^{-1}}$ is 10^4 $\mathrm{cm^2\,s^{-1}}$.

2.1 Dai et al. (2024): the model this project uses

2.1.1 Objective and Methodology

Dai, Shao and Sheng built VULCAN-Venus, an open-access one-dimensional photochemistry and transport model, to study how the chemistry depends on mixing in the clouds and on the loss of SO_2 into the droplets [4]. It spans the surface to 112 km in 2 km layers, with 362 neutral and 45 photolysis reactions among 59 species, mainly from Mills (1998) and Krasnopolsky (2007) (as cited in [4]). Sunlight is followed from 100 to 700 nm, and the droplets scatter it in two size modes. The absorber is not a chemical species: its absorption between 320 and 560 nm is attached to the mode 1 droplets [4]. The nominal K_{zz}, from Bierson and Zhang (Section 2.3), rises from about 17 000 $\mathrm{cm^2\,s^{-1}}$ at the surface to about 53 000 at 80 km and steeply above; a case called Model A replaces it between 46.5 and 55 km with a

stronger profile from Venus Express data. Three reactions taken from Rimmer et al. (Section 2.4) move SO_2 into and out of the droplets as liquid SO_2, with rates from Henry's law rather than measurement [4].

2.1.2 Results and Critique

The model reproduces the observed fall of SO_2 through the clouds: SO_2 dissolves between 58 and 70 km and is released between 40 and 58 km, where the droplets evaporate [4]. Model A changes the chemistry only slightly. OSSO reaches only about 10^{-8} ppmv at 70 km, which the authors judge insufficient for the absorber; its photolysis is omitted for lack of laboratory data [4].

This project's reading. The absorber's amount, height and spectrum are inputs, so the published model cannot test whether a candidate supplies them; making the absorber switchable was the first task of this project (Chapter 3). The authors call the liquid SO_2 a "substitute medium" tuned to the observations [4], so SO_2 above the clouds is fitted, not predicted. The mixing test stops at 55 km, below the cloud top where the absorber acts. And although the abstract speaks of mixing retrieved from Venus Express, the nominal run uses the Bierson and Zhang profile.

2.2 Pérez-Hoyos et al. (2018): the absorber's spectrum and height

2.2.1 Objective and Methodology

Pérez-Hoyos and co-workers analysed 318 spectra from the MASCS spectrometer on MESSENGER, taken on 5 June 2007 along the equator from near noon to the evening terminator [13]. A radiative transfer retrieval gave the height and amount of four modes of cloud particles and the absorber's spectral shape. The absorber was assumed well mixed with mode 1, and only the imaginary part of the mode 1 refractive index, the part that measures absorption, was retrieved, with SO_2 fixed at 500 ppb. Candidates were then ranked by how closely their spectra follow the result.

2.2.2 Results and Critique

The cloud top, where the optical depth at 0.63 µm reaches one, is at 75 ± 2 km; mode 1 starts at 65 ± 2 km with a scale height of about 4 km. The absorption is centred at 0.34 ± 0.03 µm with a full width at half maximum of 0.14 ± 0.01 µm, if

Gaussian, and the data are most sensitive to it at 75 ± 7 km [13]. In the ranking, S_2O fits best and OSSO next, whose shape the authors find closer; $FeCl_3$, SCl_2 and Cl_2 absorb over too narrow a band [13]. These values are the observed targets used in this report (Figure 1.2).

This project's reading. The authors ask a candidate for the right spectrum, a number density consistent with photochemical models, and a lifetime consistent with the motion of the ultraviolet markings, but test only the spectrum, as they state [13]. The absorber's height is tied to mode 1 by assumption and cannot be retrieved separately. The paper also calls the band shifted to longer wavelengths than an earlier estimate in its results but to shorter ones in its conclusions; this report uses only the centre and width.

2.3 Bierson and Zhang (2020): mixing and the SO_2 problem

2.3.1 Objective and Methodology

Bierson and Zhang built the first one-dimensional model to join the lower and middle atmosphere in one domain, from the surface to 112 km, as a tool for studying how species and regions interact rather than a tuned fit [2]. The absorber is an imposed opacity, and water and sulfuric acid vapour are fixed because no clouds form. They compared four K_{zz} profiles: the nominal one, extended downward from an earlier middle-atmosphere model; one from Krasnopolsky; the nominal divided by ten (Lowkzz); and the nominal lowered to 1000 $cm^2\,s^{-1}$ between 45 and 65 km (Cloud kzz) [2]. VULCAN-Venus adopts the nominal profile, as does every Phase 1 run (file `atm/atm_Venus_Bkzz.txt`).

2.3.2 Results and Critique

With the nominal profile SO_2 stays almost uniformly mixed and does not fall above the clouds [2]. Lowkzz and Cloud kzz come close, but the authors call Lowkzz instructive for understanding the model's sensitivity rather than plausible, and note that measured and estimated mixing in the clouds is at least three times stronger than either. They conclude that "some as of yet unknown chemical sink in the cloud region or interactions with the cloud droplets" is needed [2]. OSSO, written $(SO)_2$, is below 10^{-25} at 58 km, never exceeds 1 ppb, and peaks near 90 km, far above the cloud top. About 40 per cent of the rate coefficients have no laboratory value [2].

This project's reading. Phase 1 uses a profile whose authors show it cannot reproduce

SO_2 without an extra sink in the clouds, which Dai et al. supply. The profile is published only as a figure, so its values come from the model's input file. The verdict on OSSO rests on abundance and height, not spectrum.

2.4 Rimmer et al. (2021): chemistry inside the droplets

2.4.1 Objective and Methodology

Rimmer and co-workers tested three explanations for the loss of SO_2 and water in and above the clouds: more water below the clouds, less SO_2 below them, and chemistry inside the droplets enabled by dissolved hydroxide salts, with sodium hydroxide as a stand-in [14]. In their 480 species model, gas dissolves into the droplets by Henry's law and the salts neutralise part of the acid so that more SO_2 is held. The absorber is a fixed optical depth between 58 and 67 km, fading above.

2.4.2 Results and Critique

Without extra hydrogen, the water available removes only about 20 per cent of the SO_2 [14]. More water fails, raising the cloud top 20 km too high and the oxygen far too much. Less SO_2 below the clouds works at 15 to 20 ppm (the paper gives both), against about 150 ppm observed. Salts reproduce SO_2 with a droplet pH of about 1 at 50 km, but need a continuous supply of about 10^{-13} mol cm^{-2} s^{-1}, some four orders of magnitude more than delivery from space [14]. No gas in their models explains the absorber above 200 nm, and the particles' own absorption was not considered.

This project's reading. The salt profile is prescribed at each height to fit SO_2, its source lies outside the mass balance, and the authors call the solution not unique [14]. The paper shows that buffered droplets *could* hold the missing SO_2, not that a real salt reaches them. Dai et al. adopted its droplet reaction rates.

2.5 Lefèvre et al. (2022, 2024): resolved convection in the clouds

2.5.1 Objective and Methodology

Lefèvre and co-workers measured K_{zz} instead of assuming it, with a large-eddy simulation: a three-dimensional model fine enough to resolve single rising plumes, with a 400 m grid over a box 64 km wide, from the surface to 90 km [10]. The 2022

study followed idealised tracers for SO_2 and water at the equator and at 75 degrees latitude at noon, and derived K_{zz} from their simulated vertical fluxes. The 2024 study added full photochemistry, equilibrium clouds and midnight, and derived an effective K_{zz} for each species [11]. The solar heating corresponds to a fixed, average amount of absorber [10].

2.5.2 Results and Critique

A deep convective layer sits at about 47 to 56 km at the equator. There, at noon, a cloud-top convective layer appears at about 66 to 74 km, driven by the absorber's own solar heating; it is absent at night and at high latitude, and the authors call it still speculative [10]. The effective K_{zz} in the deep layer is about 10^3 $m^2\,s^{-1}$ for SO_2 and 10^2 for water; in the cloud-top layer it rises from about 1 $m^2\,s^{-1}$ at 65 km to 10^4 at 72 km, raising SO_2 at 72 km from 0.17 to 0.7 ppm [11]. These values are orders of magnitude above those in one-dimensional chemistry models [11], none of which yet includes the cloud-top convection [10].

This project's reading. Neither paper moves or changes the absorber, though the 2022 paper says its uncertain abundance "could have a strong effect" on the cloud-top convection [10]. Absorber and mixing are coupled: the heating drives mixing that redistributes the absorber. Neither paper reports mixing above about 75 km, leaving most of this project's 69 to 93 km response band (Chapter 5) unconstrained, and the species-dependent K_{zz} shows that VULCAN-Venus's single K_{zz} is a simplification. These points motivate Phase 2 (Chapter 9).

2.6 Jiang et al. (2024): iron sulfate minerals

2.6.1 Objective and Methodology

Jiang and co-workers tested in the laboratory which iron sulfate minerals survive in sulfuric acid of droplet strength, and whether their absorption matches the absorber [9]. Minerals were kept in 50 to 98 weight per cent acid for up to 30 days at room temperature and 1 bar and followed by Raman spectroscopy. The absorbance of the surviving minerals and of dissolved iron, normalised to a maximum of one between 200 and 500 nm, was fitted to the observed spectrum.

2.6.2 Results and Critique

Rhomboclase is stable up to 74.2 wt% acid and becomes acid ferric sulfate in stronger acid: within 14 days at 79.1 wt%, two hours at 83.5 wt% and five minutes at

97.4 wt% [9]. The best fit mixes 1 wt% rhomboclase in 74.2 wt% acid with 1.25 wt% acid ferric sulfate in 83.5 wt% acid; dissolved iron matches 200 to 300 nm. The authors estimate that about 1 to 2 wt% of absorber is needed, against at most 10^{-3} wt% iron from generous estimates of delivery from space, and that rhomboclase forms only where the droplets are out of equilibrium or in balance with about 100 ppm or more of water vapour, above the 30 ppm observed below the clouds [9].

This project's reading. Normalised spectra test shape, not strength; the absolute coefficients are in the supplementary material, which Experiment 13 uses (Chapter 6). Nothing is modelled about transport, settling or chemistry, so whether the minerals stay at the cloud top, and how the iron arrives, is untested. Dai et al. note that the minerals formed under a particular laboratory setting, not a Venus-like one [4].

2.7 Trabelsi and Francisco (2026): chlorine and sulfur molecules

2.7.1 Objective and Methodology

Trabelsi and Francisco calculated from first principles the structure, vibrations, excited states and cross sections from 200 to 600 nm of three forms of S_2Cl_2 (ClSSCl, $SSCl_2$ and a ring form), as isolated gas molecules at absolute zero [17].

2.7.2 Results and Critique

ClSSCl absorbs strongly near 240 nm, against a measured band near 256 to 260 nm. $SSCl_2$ absorbs from about 410 nm, with maxima near 348 and 262 nm, and since this overlaps the absorber's range the authors conclude it "could contribute" [17]. They offer the results as better inputs for photochemical models.

This project's reading. The paper gives spectra, not amounts: no abundance, production, lifetime or altitude. A molecule that absorbs in the near ultraviolet is also broken up by it, so it would need a continuous source. The authors supplied their cross sections, which Experiment 11 runs in the model (Chapter 6).

2.8 Egan et al. (2025): OSSO tested end to end

2.8.1 Objective and Methodology

Egan and co-workers asked whether OSSO can be abundant enough [6]. They computed new rate coefficients for its formation from two SO molecules and for its loss,

placed them in a three-dimensional model of the whole atmosphere with a sulfur source from meteors, ran it for nine Venus days to a steady state, and compared the resulting reflectance, from a radiative transfer model, with the MESSENGER spectrum.

2.8.2 Results and Critique

OSSO forms a thin layer near 70 km, about three orders of magnitude short of the observed absorption, still two orders short with its loss to oxygen and chlorine atoms switched off, and with too narrow a band [6]. SO_2 above the clouds was kept deliberately too high, which favours OSSO, and the authors conclude that OSSO does not contribute significantly.

This project's reading. This is the only peer-reviewed paper reviewed here in which the chemistry sets a candidate's amount and the reflectance follows from it, and so the only independent check on this project's screen, which reaches the same verdict against a different yardstick (Experiment 12, Chapter 6). Its radiative step uses one column at the equator near midday, and its loss rates are calculated, not measured.

2.9 Spacek et al. (2026): how strongly the droplets must absorb

2.9.1 Objective and Methodology

Spacek, Rimmer, Petkowski and Lee asked how strongly the droplet liquid must absorb, in units a laboratory measures [15]. They fitted a multiple scattering model to dayside reflectance between 365 and 455 nm from a 2020 campaign, with the absorber spread evenly through mode 1 and mode 2 droplets between 59 and 65 km and represented by a lowered *single scattering albedo*, the fraction of light meeting a particle that is scattered rather than absorbed. The absorption per unit droplet volume gives the *decadic absorption coefficient* a of the liquid: the absorbance of a 1 cm path in a laboratory cell.

2.9.2 Results and Critique

The single scattering albedo must be 0.89 at 365 nm, and a peaks at 1278 cm^{-1} at 375 nm, falling below 54 cm^{-1} at 455 nm [15]. An efficient organic absorber would have to make up about 0.5 to 7 per cent of the droplet liquid, nearly four orders of magnitude above the authors' estimate of the organic carbon available; weaker

inorganic absorbers would have to be a major part of the droplets. Ferric chloride is rejected because 1 g per litre of *water* gives about 1 cm^{-1} at 375 nm, so a straight-line extrapolation needs about 1.3 kg per litre [15].

This project's reading. The requirement holds only for the assumed layer, mixing and droplet sizes, as the authors acknowledge. The ferric chloride figure rests on a measurement in water, though the same paper warns that aqueous spectra have limited value unless Venus conditions are reproduced [15], and it cites the sulfuric acid measurements of Egan et al. only for spectral shape. Experiment 14, done after Phase 1 closed, repeats the calculation with the same group's measurements of ferric chloride in hydrochloric acid (Section 2.10 and Chapter 6).

2.10 Egan et al. (2025, 2026): ferric chloride in sulfuric acid

2.10.1 Objective and Methodology

The ferric chloride spectrum used in recent comparisons had been measured in an organic solvent with which it can react [7]. Egan, James and Plane measured it from 200 to 600 nm in about 77 and 78 wt% sulfuric acid with added hydrochloric acid, at room temperature, and followed its slow conversion to ferric sulfate for about a hundred days [7]. To fit these mixtures they also derived its molar absorptivity in hydrochloric acid of several strengths (their Figure 5a), the curves Experiment 14 uses [7]. A preprint by the same group, posted on 14 January 2026 and marked as not peer reviewed [8], added 57 iron reactions to a three-dimensional model of Venus, with iron from meteors and chlorine from atmospheric hydrogen chloride, and used the laboratory spectra in a radiative transfer model compared with the MESSENGER spectrum.

2.10.2 Results and Critique

Ferric chloride had not been measured in concentrated sulfuric acid before, the solvent has a significant effect on its spectrum, and hydrochloric acid slows its conversion to sulfate [7]. Its normalised shape agrees well with the absorber, but the authors add that altitude, formation, lifetime and transport "must all be considered through extensive atmospheric modeling" first. The preprint finds that 1.0 to 1.2 wt% ferric chloride in mode 1 droplets reproduces the absorption between 300 and 600 nm (1.2 to 1.5 wt% if part turns to sulfate), and that a purely meteoric source needs the iron to stay in the clouds for about 3000 years [8].

This project's reading. The laboratory spectra were taken several hundred to about a thousand times more dilute than the fitted loadings, so using them at those loadings assumes that absorbance grows in proportion to the amount. In the preprint the amount comes from the radiative fit, not the chemistry, whose run had not reached a steady state, and the residence time is 2800 and 3400 years in the discussion but 1800 to 3000 in the conclusions [8]. Most important here, Spacek et al. and the preprint reach opposite conclusions on ferric chloride and neither cites the other. Experiment 14 (Chapter 6) traces the disagreement to the solvent of the input measurement; it does not identify the absorber.

2.11 What the literature leaves open

Against the three tests of Section 1.4, the coverage is uneven. **The spectrum** is tested most often, almost always as a normalised shape. **The amount** has been tested for a few candidates, one at a time, and only Spacek et al. state the absolute strength the droplets need. **The altitude** is assumed: tied to mode 1, set as a fixed layer, or prescribed. No paper reviewed lets a chemistry model place a candidate by itself and then measures how the radiation responds to where it sits, while the mixing that would carry a candidate there is uncertain by orders of magnitude and may be driven, at the cloud top, by the absorber itself. This is the gap of Section 1.5, and the following chapters describe how Phase 1 addressed it.

Chapter 3

The Model and the Groundwork

Before any experiment could be run, three things had to exist: a model whose working was understood, a baseline run that could be trusted, and a way to switch the absorber on and off while changing nothing else. This chapter explains how a photochemistry and transport model works (Section 3.1), describes VULCAN-Venus as it is used here (Section 3.2), and then tells the groundwork in the order it happened: the environment and baseline (Section 3.3), a false lead (Section 3.4), the discovery that made controlled experiments possible (Section 3.5), and the requirement every candidate is later measured against (Section 3.6). Section 3.7 shows how the experiments grew from it.

3.1 How a photochemistry and transport model works

The model treats the atmosphere as a column of horizontal layers, the same everywhere around the planet, so that only height varies; this is what *one-dimensional* means. In each layer it follows the number density n_i of every species i, the number of its molecules per cubic centimetre. The density changes for two reasons: chemistry makes or destroys the species, and mixing carries it between layers. The model solves, for every species and every layer,

$$\frac{\partial n_i}{\partial t} = P_i - L_i - \frac{\partial \phi_i}{\partial z}, \tag{3.1}$$

where t is time, z is altitude, P_i and L_i are the rates at which reactions produce and destroy species i (molecules per cubic centimetre per second), and ϕ_i is its upward flux (molecules per square centimetre per second). The flux is given by eddy diffusion,

$$\phi_i = -K_{zz}\, n\, \frac{\partial X_i}{\partial z}, \tag{3.2}$$

where n is the total number density of the air, $X_i = n_i/n$ is the mixing ratio of species i, and K_{zz} is the eddy diffusion coefficient of Chapter 2. The minus sign makes a species flow from where its mixing ratio is high to where it is low.

Many reactions are driven by sunlight. A molecule that absorbs a photon of enough energy can break apart; this is *photolysis*. How much sunlight reaches a layer depends on how much has been absorbed or scattered above it, measured by the optical depth. At the lower boundary of layer j and wavelength λ the model adds up that layer and every layer above it:

$$\tau(j,\lambda) = \sum_{k \geq j} \Big[\sum_{s} n_s(k)\, \sigma_s(\lambda) \Big]\, \Delta z_k, \tag{3.3}$$

where $n_s(k)$ is the number density of absorbing or scattering species s in layer k, $\sigma_s(\lambda)$ is its cross section, the effective area one molecule presents to light (square centimetres), and Δz_k is the layer thickness in centimetres. Sunlight arriving from a direction at angle θ to the vertical is weakened in proportion to $e^{-\tau/\cos\theta}$. The model then computes the *actinic flux* $F(z,\lambda)$, the number of photons per square centimetre per second per nanometre arriving at a point from all directions, including light scattered by the gas. The rate coefficient of a photolysis reaction of species s is

$$J_s(z) = \int \sigma_s(\lambda)\, q_s(\lambda)\, F(z,\lambda)\, \mathrm{d}\lambda, \tag{3.4}$$

where $q_s(\lambda)$ is the fraction of absorbed photons that break the molecule, so that J_s is the probability per second that one molecule is broken up. The destruction rate it contributes to L_s is $J_s n_s$, halved in this model for day and night (Table 3.1).

These equations are coupled. An absorber in a high layer raises τ below it, which lowers F, which lowers every J, which changes the chemistry, which changes the abundances that set τ. This chain, from optical depth to actinic flux to photolysis to chemistry, is the path by which the unknown absorber can affect the model at all.

The model is integrated in time until it no longer changes, which is called *steady state*. VULCAN-Venus declares steady state when three tests pass at once: the largest fractional change of any mixing ratio over the recent steps, called `longdy`, is below 0.01; that change divided by the elapsed time, `longdydt`, is below 10^{-5} per second; and the largest fractional change of the actinic flux from one step to the next is below 0.1 (`yconv_cri`, `slope_cri` and `flux_cri` in `vulcan_cfg.py`, tested in `op.py`).

3.2 VULCAN-Venus as used in this project

VULCAN is a general photochemistry and transport model; VULCAN-Venus is its adaptation to Venus by Dai, Shao and Sheng [4], reviewed in Section 2.1. Table 3.1

lists the settings of every Phase 1 run.

Table 3.1: The model set-up used in every Phase 1 run. Configuration lines are in `vulcan_cfg.py`; counts from saved output are in the registry under the keys shown.

Property	Value	Source
Vertical grid	0 to 112 km in 57 layers: 56 layers 2 km thick and a vanishingly thin layer at the surface	`atm/alt_venus.txt`
Temperature and pressure	733 K and about 92 bar at the surface (the model's own profile, a little cooler than the 750 K of Chapter 1); about one millionth of a bar at the top	`atm/atm_Venus_Bkzz.txt`
Eddy diffusion	Bierson and Zhang nominal profile [2]	`atm/atm_Venus_Bkzz.txt`
Chemistry	Network `Nominal.txt`: 62 entries in the species list, 838 reaction slots (a forward and a reverse slot per reaction)	`nom.base.v.nr`
Wavelength grid	1670 bins, 96 to 698 nm; 0.1 nm wide below 240 nm, 2 nm above	`nom.base.v.nbin`, `.dbin1`, `.dbin2`
Sunlight	One zenith angle of 58 degrees, the dayside average; every photolysis rate halved for the day and night average	`sl_angle`, `f_diurnal`
Fixed surface mixing ratios	SO_2 100 ppmv, COS 30 ppmv, CO 15 ppmv, among others	`use_fix_sp_bot`
Fixed profiles	H_2SO_4 vapour and liquid, water vapour and condensed water, N_2, and the two particle tracers	`fix_species`
Switched off	Vertical wind, molecular diffusion	`use_vz`, `use_moldiff`
Absorber input	$Q_{\mathrm{abs}}(\lambda)$ table; mode 1 particle profile	`atm/UV_absorber.txt`, `atm/mode1+2.txt`

Three features of this set-up matter for what follows. First, the column is a single global average at one sun angle, so day and night, latitude and season are not represented. Second, the sulfuric acid and water that make up the clouds are fixed profiles; the model does not form or move cloud droplets. Third, as Section 3.5 explains, the radiation in the code as supplied included gas absorption, scattering by molecules and the cloud droplets in a simplified form, but not the absorber.

3.3 The environment and the baseline

Work began by building a working environment, Ubuntu 24.04 running under WSL2 on Windows with a Conda environment named `venus`, as recorded in `documentation/SETUP_AND_REPRODUCTION.md`. The supplied code predates the current Python libraries, and two small compatibility corrections were needed: one line in `make_chem_funs.py` that assumed text arrives as raw bytes, and one import in the PyMieScatt library whose function had been renamed in SciPy. Neither changes any chemical, atmospheric or optical input.

The order of the next two steps mattered. The output file supplied with the model was first copied and checked to be identical, byte for byte, to the original; this copy is called the *reference*. Only then was the model run afresh, giving the *baseline*. Without a preserved reference, no later disagreement could have been pinned on anything. Both runs met the steady-state tests of Section 3.1. The baseline needed 296 steps and reached a model time of 5.328×10^8 s, with `longdy` equal to 0.0099016 and `longdydt` equal to 6.971×10^{-11} s^{-1}; the reference had been integrated much longer, over 14336 steps to 3.082×10^{10} s, and ended at 0.0097880 and 7.219×10^{-11} s^{-1} (registry keys beginning `nom.base` and `nom.ref`).

The two states were then compared quantity by quantity: mixing ratios, optical depth, actinic flux and photolysis rates. The grid, pressure, temperature, layer thickness, K_{zz} and cross sections were identical. The states agreed closely almost everywhere, with one exception.

3.4 A false lead: the 200 to 220 nm anomaly

Around 54 to 58 km and between about 200 and 220 nm, the optical depth of the two states differed sharply. At 54 km (grid interface 28) and 201.6 nm, the reference optical depth is 1376.448 and the baseline 933.950, a difference of 442.498 (registry keys beginning `falselead`).

This looked, briefly, like a signal: it was in the ultraviolet and inside the clouds. It would have been easy to take it as a trace of the absorber. It was traced instead. Since all inputs were identical, the difference had to come from the computed composition. In layer 28 the SO_2 mixing ratio differs by about a third between the two states, and recomputing Equation (3.3) from the two SO_2 profiles alone reproduces essentially the whole optical depth difference at that point. Going one step further back, gas-phase sulfuric acid and SO_3 in that layer are about fourteen times lower in the baseline, while the fixed liquid sulfuric acid is identical. The reference had been integrated nearly sixty times longer than the baseline. The chain therefore runs from a difference in sulfur chemistry between the two states to SO_2, and from SO_2 to optical depth, where SO_2 absorbs strongly.

This conclusion was reached before any experiment was run: the 200 to 220 nm anomaly is a difference between two model states caused by ordinary SO_2 absorption, and it must not be used as evidence of the absorber. Rejecting it kept the control of every later experiment clean.

3.5 The discovery: an absorber term that was switched off

The absorber was then audited directly in the source code. The code reads the absorber table, a list of absorption efficiencies $Q_{\mathrm{abs}}(\lambda)$, interpolates it onto the model's wavelength grid, takes the absorbing particles to be the mode 1 droplets with diameter $D = 0.98$ µm, and takes their number density from the mode 1 profile. From these it forms, for each layer j, the absorber's share of the optical depth,

$$\tau_{\mathrm{UUV}}(j, \lambda) = Q_{\mathrm{abs}}(\lambda)\, \frac{\pi D^2}{4}\, N(j)\, \Delta z_j, \tag{3.5}$$

where $\pi D^2/4$ is the cross-sectional area of one particle, $N(j)$ is the number of particles per cubic centimetre in layer j, and Δz_j is the layer thickness in centimetres. Q_{abs} is the fraction of light falling on that area that the particle absorbs. The subscript UUV, for *unknown ultraviolet*, is the name the code uses.

The decisive finding was that this calculation, together with a full scattering calculation for the two modes of cloud droplets, sat inside a block of code that had been turned into a comment, in the function `compute_tau()` of `op.py`. It was present but never executed. The comment beside it says why: the droplet calculation had been replaced by fixed, fitted cross sections for the two droplet modes, carried as two particle species named `V` and `V2`, to save computation. The optical depth the supplied code actually computed was therefore gas absorption, scattering by molecules and these two droplet species, and the saved output held no absorber arrays. Recomputing the saved optical depth from those terms matched it to very small numerical residuals.

This is what made the project possible. Because the term existed, it could be switched on under a single number, the amplitude A_{UUV}:

$$\tau^{*}_{\mathrm{UUV}} = A_{\mathrm{UUV}}\ \tau_{\mathrm{UUV}}. \tag{3.6}$$

With $A_{\mathrm{UUV}} = 0$ the model is exactly as supplied, which is the *control*. With $A_{\mathrm{UUV}} = 1$ it carries the supplied absorber exactly as tabulated, which is the *treatment*. The two runs differ in one term and in nothing else, so any difference between them can be blamed on the absorber alone, through the chain of Section 3.1. Only this term was switched on; the full droplet calculation stayed off, and the droplets enter every run, control and treatment alike, through the same fitted cross sections, so that the comparison isolates the absorber. The source

file was copied before the change and the copy checked identical by its SHA-256 hash (`op-25m0328-pre-UUV-experiment.py`), and the amplitude is set in `vulcan_cfg.py` as `A_UUV`.

The absorber as supplied. The absorber that comes with the model, called the *empirical absorber* in this report, has two parts, and every later experiment changes one or both of them. Its *spectrum* is the table of Q_{abs} in `atm/UV_absorber.txt`. It is non-zero from 302 to 698 nm (`spec_emp.sup_lo` and `.sup_hi`), largest at 320 nm, where it is 0.356 (`spec_emp.peakQ`), and 61.58% of its absorption lies between 320 and 400 nm (`share320400.emp`). Its *profile* is the mode 1 particle density in `atm/mode1+2.txt`. The density is largest at 60 km, with 595 particles per cubic centimetre (`pnom.peak_z` and `pnom.peak_N`); the profile is centred at 63.419 km (`pnom.centroid`), holds 90.194% of its particles between 58 and 69 km (`pnom.p5869`) and none below 30 km, and adds up to 4.0776×10^{8} particles per square centimetre over the whole column (`pnom.column`). Experiment 7 moves the profile and keeps the spectrum; Experiments 8 and 11 change the spectrum and keep the profile; Experiments 9 and 10 change both.

3.6 The requirement constant

Adding Equation (3.5) over every layer gives the absorber's optical depth through the whole atmosphere, which is the amount of absorption any real candidate must supply. It is largest at 320 nm, where the table peaks:

$$\tau_{\mathrm{req}} = Q_{\mathrm{abs}}(320\ \mathrm{nm})\,\frac{\pi D^{2}}{4}\sum_{j} N(j)\,\Delta z_{j} = 1.094968472081. \tag{3.7}$$

Here $Q_{\mathrm{abs}}(320\ \mathrm{nm}) = 0.356$ (registry key `spec_emp.peakQ`), the particle area is 7.543×10^{-9} cm^{2} (`const.area`), and the column of mode 1 particles is 4.0776×10^{8} per square centimetre (`pnom.column`). The result is held as `const.tau_req`. It is the same in the reference and the baseline, as it must be, because it depends only on input tables.

In plain terms, sunlight at 320 nm that passes straight down through the absorber layer keeps a fraction $e^{-\tau_{\mathrm{req}}}$ of its intensity, about one third. Because this number was recovered from the model's own inputs rather than taken from a paper, every candidate in Chapter 6 is measured against exactly the absorption the model's own absorber supplies. When the model was run with the absorber switched on, its optical depth at 56 km, below the absorber, exceeded that of the control by this same amount at 320 nm, as expected.

3.7 From groundwork to experiments

With a trusted baseline and a clean control, the experiments followed in four stages, shown in Figure 3.1. Experiments 1 and 2 switch the absorber on and ask whether, and what, it changes. Experiments 3 to 6 locate the response in amount, wavelength, reaction and altitude. Experiments 7 to 10 ask what decides the size of the response, and Experiments 11 to 14 test real candidates against what the earlier stages found. Chapter 4 first states the conventions used for every number, and Chapters 5 and 6 describe the experiments in the order they were done.

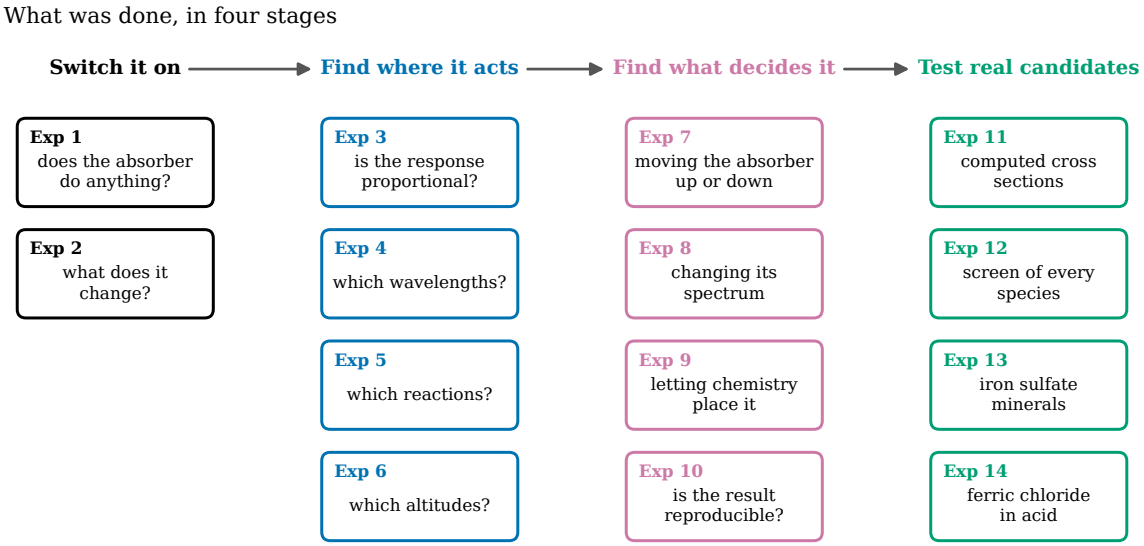


Figure 3.1: What was done, in four stages. Each box is one experiment and the question it asked, and each stage builds on what the stage before it found. Source: `figures/scripts/F04_CTX_what-was-done.py`.

Chapter 4

Method and Conventions

Every experiment in Chapters 5 and 6 compares two converged model states and reports how they differ. A difference can be measured in more than one way, and two honest calculations from the same output can give different numbers if they make different choices. This chapter states every choice that was made, so that a reader who repeats a calculation gets the number printed here: how numbers are written (Section 4.1), the six conventions with a worked example each (Section 4.2), the fixed ranges used throughout (Section 4.3), how the size of a change in the whole radiation field is measured (Section 4.4), the response band (Section 4.5) and the reproducibility floor (Section 4.6).

The same six conventions are also written out in the code that computes every number, `verify/recompute_25m0328.py`: the first five at the top of the file, and the sixth as the value of the key `res.e10.SO.inband`. Where a recomputation disagrees with a number in this report, the first thing to check is whether it follows these conventions.

4.1 How numbers are written in this report

The registry, the file `verify/registry.json`, holds every quantity computed from the model, each under its own name, called a *registry key*; an example is `e6.A1.meanz`. Chapter 8 explains how to read these names. Where this report prints such a number, the key is named next to it or at the start of the paragraph.

A printed number carries only as many digits as the text needs. It is correct when the registry value, rounded to the same number of significant digits, gives the printed number. For example, `e6.A1.meanz` holds the response centroid to sixteen digits, and the text prints it as 82.044 km. The project's checker, `verify/check_documents_25m0328.py`, applies this rule against the whole registry to every number written with three or more decimal places, in scientific notation, or as a percentage with two or more decimal places. Percentages may be stored either as percentages or as fractions; 1.4484% and a stored 0.014484 are the

same number.

The checker can show that a printed number exists in the registry. It cannot show that it was taken from the right key. That second check is done by hand, by recomputing the sentence from the key it names, and Chapter 8 gives an example where it mattered.

4.2 The six conventions

Table 4.1 lists the six conventions. Each is explained below with a worked example.

Table 4.1: The six conventions used for every number in this report.

No.	Convention	What it means
1	Relative change	The difference is divided by the larger of the two values, not by the control.
2	Means over a field	Every element counts equally. A *bounded* mean leaves out the elements where both values are zero.
3	Integrals over wavelength	VULCAN's own rule: two bin widths, and half weight at the two ends of every range.
4	Integrals over altitude	Layer thickness in centimetres, never kilometres.
5	Two photolysis measures	ΔJ (the rate coefficient) and $\Delta(Jn)$ (the rate) are different quantities and are never mixed.
6	Reproducibility floor	In-band changes in abundance smaller than 2.5278% are not resolved.

4.2.1 Relative change divides by the larger value

When a quantity X has the value X_{r} in the reference run and X_{c} in the case being tested, its relative change is

$$R_X = \frac{|X_{\mathrm{c}} - X_{\mathrm{r}}|}{\max\left(|X_{\mathrm{r}}|,\, |X_{\mathrm{c}}|\right)}. \tag{4.1}$$

The denominator is the larger of the two magnitudes. This keeps R_X between 0 and 1, or between 0 and 100 per cent, whatever the two values are, and it is defined even when the reference value is zero. For a small change the larger value and the reference value are almost equal, and the two possible definitions agree. They differ only when the change is large, which is exactly where a percentage needs care.

Worked example. In Experiment 2 the mixing ratio of OSSO (S_2O_2) at 33 km rises from 4.012×10^{-42} with the absorber off to 3.0845×10^{-41} with it on (registry keys `e2.S2O2.ref_at_level` and `e2.S2O2.case_at_level`). The change, 2.683×10^{-41}, divided by the larger value gives $R = 0.8699$ (`e2.S2O2.maxrel`). Divided by the control instead it would give about 669 per cent, a number that says

more about how tiny the control value is than about the change. A second case shows why the rule is needed at all: ClS_2 at 97 km is exactly zero with the absorber off and 6.9437×10^{-27} with it on (`e2.ClS2.case_at_level`). A change relative to the control is then undefined, while Equation (4.1) gives exactly 1.

Both values in this example are far below anything that could be observed. That is why Chapters 5 and 6 always give an absolute value next to a relative change for trace species.

4.2.2 Means over a field

The model saves its fields as arrays: the actinic flux, for example, has one value for each of 57 layers and 1670 wavelength bins, 95190 values in all. Where a mean is reported over such an array, every element counts equally; no element is weighted by layer thickness or bin width. The one exception is the *bounded* mean of Equation (4.1), which is taken only over elements where the denominator is not zero:

$$\overline{R} = \frac{1}{M} \sum_{\text{elements with } \max(|X_\mathrm{r}|,|X_\mathrm{c}|)>0} R_X, \tag{4.2}$$

where M is the number of those elements. An element where both runs give exactly zero has no change to report, and counting it as a zero would dilute the mean by an amount that depends only on how much of the grid is dark.

Worked example. Between the runs with the absorber off and on (Experiments 1 and 2), the actinic flux is non-zero in only 37185 of its 95190 elements; the rest are short wavelengths that are absorbed completely before they reach that layer, and the lower the layer, the more of the spectrum is dark (`e3fp.bnd.A1.aflux.n`). The bounded mean relative change over the 37185 lit elements is 3.7078% (`e3fp.bnd.A1.aflux.mean_pct`). The mean over all 95190 elements, with each dark element counted as zero, is 1.4484% (`e2.aflux.meanrel`). The two differ by exactly the share of lit elements, and both are correct for what they say. This report states which one it uses.

4.2.3 Integrals over wavelength

The photolysis rate coefficient of Equation (3.4) is an integral over wavelength, which the model evaluates as a sum over its bins. The bins are 0.1 nm wide from 96 to just below 240 nm (the first 1440 bins) and 2 nm wide from 240 to 698 nm (`nom.base.v.dbin1` and `.dbin2`). VULCAN adds up each of the two parts with the trapezoidal rule, which gives the two end bins of each part half weight. For a

range of bins from a to b with width $\Delta\lambda$,

$$\int_a^b f(\lambda)\,\mathrm{d}\lambda \approx \sum_{i=a}^{b} f_i\,\Delta\lambda \;-\; \tfrac{1}{2}\big(f_a + f_b\big)\,\Delta\lambda, \tag{4.3}$$

where f_i is the integrand in bin i. Every band integral in this report, for example the share of a response that falls between 320 and 340 nm, uses the same rule at its own two edges.

Worked example. A band from 320 to 340 nm holds 11 bins of 2 nm, centred at 320, 322, and so on up to 340 nm. For an integrand equal to 1 the true integral is 20 nm. A plain sum of the 11 bins gives 22 nm, a tenth too much; Equation (4.3) removes half a bin at each end and gives 20 nm. With this rule, recomputing the stored photolysis rate coefficients from the stored actinic flux and cross sections gives back the model's own values: the difference is exactly zero in 102 of the 110 checks made (55 photolysis branches in two runs), and very small in the rest (registry keys ending `recon_maxerr`).

4.2.4 Integrals over altitude

Many model quantities are per cubic centimetre: number densities, and reaction rates in molecules per cubic centimetre per second. Integrating such a quantity over altitude gives a column, per square centimetre, only if each layer thickness is in centimetres:

$$C = \sum_j r_j\,\Delta z_j, \qquad \Delta z_j \text{ in cm}. \tag{4.4}$$

A thickness of 2 km is 2×10^5 cm, so taking it in kilometres makes the result exactly 10^5 times too small and leaves its unit meaningless.

Worked example. In Experiment 7, moving the absorber upward changes the contribution that the photolysis of SO_2 (reaction 781 of the network) makes to the SO_2 budget. Integrated over the response band with the thickness in centimetres, the change is $+8.2352 \times 10^8$ molecules per square centimetre per second (`e7.7a.SO2.R781.col`). Taking the thickness in kilometres instead gives 8235.1671, the same digits 10^5 times smaller (`e7.7a.SO2.R781.col_km`). The registry keeps both so that the size of the error can be seen; only the first is used in this report.

4.2.5 Two photolysis measures that must not be mixed

A change in photolysis can be reported in two ways. The change in the rate coefficient, ΔJ, is the change in the probability per second that one molecule is broken up. It depends only on the sunlight reaching that height, so it is a purely radiative measure. The change in the rate, $\Delta(Jn)$, where n is the number density of the molecule, is the change in how many molecules are broken up per cubic centimetre per second. It also depends on how much of the molecule is present. Because most molecules become more abundant lower down while sunlight becomes weaker, the second measure usually peaks lower in the atmosphere than the first.

Experiment 6 reports $\Delta(Jn)$. Experiments 7 to 10 report ΔJ when they describe photolysis; the change in a reaction's contribution to a budget, as in Experiments 5 and 7, is a rate and so of the $\Delta(Jn)$ kind.

Worked example. For the photolysis of SO_3 in Experiment 9 (Chapter 6), the largest change of the rate coefficient inside the response band is 0.05898%, at 81 km (`dJ.e9.SO3` and `dJz.e9.SO3`). The largest change of the rate is 0.4603%, at 73 km (`dJy.e9.SO3` and `dJyz.e9.SO3`). The two measures describe the same run and differ in size by a factor of about eight and in height by 8 km. A number taken from one and placed in a sentence about the other would look like a disagreement in altitude where there is none.

4.3 Fixed ranges used throughout

Four ranges are used again and again, and each has the same limits everywhere in `verify/recompute_25m0328.py`, so that no analysis picks its own. The *absorber grid*, 300 to 698 nm, is where the absorber term acts: the absorber table starts at 300 nm and the model grid ends at 698 nm. The *enhancement band*, 320 to 400 nm, is the near-ultraviolet feature the absorber has to supply (Chapter 1). The *screening window*, 300 to 620 nm, is the range searched for each candidate's strongest absorption in the screens of Experiments 8 and 12. The *response band*, 69 to 93 km, is where the model's radiation responds to the absorber (Section 4.5). Experiment 4 also uses narrower analysis windows, among them 420 to 620 nm; these are deliberate divisions of the spectrum for that experiment, not errors in the ranges above.

4.4 The size of a change in the radiation field

Most radiative results in Chapters 5 and 6 compare a whole actinic flux field, 57 layers by 1670 wavelength bins, with a reference field, and report the comparison as one number: the size of the change as a percentage of the size of the reference field,

$$\Delta F = 100\,\frac{\left[\sum_j \sum_i w_i \left(F_{\mathrm{c}}(j,\lambda_i) - F_{\mathrm{r}}(j,\lambda_i)\right)^2\right]^{1/2}}{\left[\sum_j \sum_i w_i\, F_{\mathrm{r}}(j,\lambda_i)^2\right]^{1/2}} \text{ per cent,} \tag{4.5}$$

where F_{c} and F_{r} are the actinic flux in the case and in the reference, j runs over every layer, i over every wavelength bin, and w_i is the width of bin i: 0.1 nm below 240 nm and 2 nm above (the array `quad` in `verify/recompute_25m0328.py`). Each layer counts once, whatever its thickness. Squaring the differences lets the largest changes dominate, and dividing by the size of the reference field turns the result into a share of the light that is there. The function `radiative()` in the same file computes it, and the registry holds it under keys ending in `L2pct`.

Worked example. Moving the absorber 18 km upward (Experiment 7A), with the absorber in its usual place as the reference, gives $\Delta F = 2.2498\%$ (`e7a.L2pct`); moving it 18 km down gives 0.3067% (`e7b.L2pct`). The spectral substitutions of Experiments 8 and 11 use the same formula with the baseline of Experiment 8 as the reference.

One quantity uses the same sums with a different denominator. The radiation floor of Section 4.6 divides the difference between two runs by the change that one of them makes, not by the reference field, so it is a share of a change, not a share of the light. This is stated wherever it is used.

4.5 The response band

The response band answers one question: at what altitudes does the model's radiation actually respond to the absorber? Experiment 6 answered it by comparing the runs with the absorber on and off. At each layer j it adds up the change in actinic flux over all wavelengths, weighting each bin by its width:

$$w_j = \sum_i \left|F_1(j,\lambda_i) - F_0(j,\lambda_i)\right| \Delta\lambda_i, \tag{4.6}$$

where F_1 and F_0 are the actinic flux with the absorber on and off. The quantity w_j is the size of the radiative response at layer j, and its distribution over altitude shows

where the response lives. It is purely radiative, so the convention of Section 4.2.5 does not arise here.

The weighted mean altitude of this distribution, the response centroid, is 82.044 km (`e6.A1.meanz`). A quarter of the response lies below 77 km, half below 81 km and three quarters below 87 km (`e6.A1.q25`, `.q50` and `.q75`). The *response band* is the narrowest continuous range of altitudes that holds at least 90 per cent of the response. It runs from 69 to 93 km and holds 91.370% of it (`e6.interval.90.lo`, `.hi` and `.contained`). Table 4.2 shows how the range would change if a different share were chosen. The lower edge stays at 69 or 71 km for every share from 68 to 95 per cent, so the band does not depend sensitively on the choice of 90.

Table 4.2: The narrowest altitude range that holds a given share of the radiative response (Experiment 6). Registry keys `e6.interval.<share>.lo`, `.hi` and `.contained`.

Share asked for	Range (km)	Width (km)	Share actually held
50 per cent	73 to 83	10	54.567%
68 per cent	71 to 85	14	68.163%
80 per cent	71 to 89	18	81.479%
90 per cent	69 to 93	24	91.370%
95 per cent	69 to 97	28	95.240%
99 per cent	65 to 103	38	99.138%

The band is a property of the atmosphere, not of the absorber's spectrum. Experiment 4 repeated the run six times with the absorber confined to six different wavelength ranges. The response centroid of those six runs lies between 81.989 and 82.100 km (`e6.B5.meanz` and `e6.B6.meanz`), within a fraction of one model layer of the full run.

The band lies above most of the absorber's own particles: 90.194% of the mode 1 profile is between 58 and 69 km (`pnom.cloud`). The response to the absorber is therefore felt mainly higher up than the absorber sits. Chapter 5 shows the altitude profile of the response in full (Experiment 6). Every candidate test from Experiment 7 onward asks how much of a candidate's effect falls inside this band.

4.6 The reproducibility floor

A model run stops when the steady-state tests of Section 3.1 pass, and those tests allow mixing ratios to keep drifting slightly. Two runs that ought to give the same answer can therefore end a little apart. The size of that disagreement sets the smallest change in abundance that can be called a result.

Experiments 9 and 10 measured it, although they were not designed to. The two

runs put two different absorber spectra on the same low vertical profile, and Chapter 6 shows why that made them radiatively identical: the difference between their actinic flux fields is only 0.0017% of the change that Experiment 9 makes (`e10.fielddiff_pct`), and that change is itself only 0.3245% of the field (`e9.L2pct`, Equation (4.5)). Any difference between their chemistry must then be numerical, not physical. Yet, measured against the same baseline, the largest change in the SO mixing ratio inside the response band is 0.7384% in one run and 2.5278% in the other (`res.e9.SO.inband` and `res.e10.SO.inband`). For SO_3 the two are 0.3488% and 1.4288% (`res.e9.SO3.inband` and `res.e10.SO3.inband`).

The rule. A change in the abundance of a species inside the response band that is smaller than 2.5278% is not a measurement. It can be reported only as an upper bound: the true change is no larger than this, and may be zero. The convergence tolerance of 0.01 in `vulcan_cfg.py` might suggest a floor of about 1 per cent; the measured floor is two and a half times larger. It comes from a single pair of runs, so it is the smallest value the floor can have, not a full description of the scatter.

What it bounds, and what it does not. The floor applies to abundances, and so also to any quantity that contains an abundance, such as the photolysis rate $\Delta(Jn)$. It does not apply to purely radiative quantities such as the actinic flux or ΔJ: the same pair of runs agrees on those to 0.0017% of the change itself, so a radiative difference well below one per cent is still resolved. Of the 45 in-band changes in the sulfur reservoirs held for Experiments 7 to 11 (keys beginning `res.`, which divide by the reference run rather than as in Convention 1), only three reach the floor: SO and SO_3 in Experiment 7A, 8.404% and 4.718%, and the SO value that defines the floor itself. The rest are upper bounds.

Every run converged. The floor is measured between converged runs, so it matters that every run met the steady-state tests of Section 3.1. Table 4.3 gives the two tested quantities for the main runs of Chapters 5 and 6; all pass, as do the control and the runs of Experiments 3 and 4B.

Table 4.3: The steady-state tests at the end of each main run, with limits 0.01 and 10^{-5} s^{-1}. Keys `conv.<run>.longdy` and `conv.<run>.longdydt`.

Run	Key	longdy	longdydt	Run	Key	longdy	longdydt
Exp. 2, on	a1	0.0099963	6.991×10^{-11}	Exp. 8C, S_3	e8c	0.0099686	7.006×10^{-11}
Exp. 7A, up	e7a	0.0098794	6.949×10^{-11}	Exp. 9	e9	0.0098368	6.916×10^{-11}
Exp. 7B, down	e7b	0.0099450	6.992×10^{-11}	Exp. 10	e10	0.0097778	6.841×10^{-11}
Baseline, 8 to 11	b8	0.0099437	6.993×10^{-11}	Exp. 11A, $SSCl_2$	e11a	0.0099988	7.029×10^{-11}
Exp. 8A, ClS_2	e8a	0.0099390	6.987×10^{-11}	Exp. 11B, ClSSCl	e11b	0.0099701	7.012×10^{-11}
Exp. 8B, OSSO	e8b	0.0098173	6.901×10^{-11}				

Chapter 5

Experiments 1 to 7

The first seven experiments ask what the absorber does inside the model, before any candidate is tested. Experiments 1 and 2 switch it on, Experiment 3 tests whether its effect scales with its amount, Experiments 4 to 6 locate the effect in wavelength, in reaction and in altitude, and Experiment 7 moves the absorber up and down. Every comparison follows the conventions of Chapter 4. Unless stated otherwise, each experiment compares the run with the absorber on ($A_{\mathrm{UUV}} = 1$) against the control ($A_{\mathrm{UUV}} = 0$).

5.1 Experiments 1 and 2: does the absorber do anything?

Experiment 1 ran the model with the absorber term present but its amplitude set to zero, and compared the result with a second run of the unchanged model, made on the project computer before the term was switched on (keys beginning `e1.ref`; it is a separate run from the baseline of Section 3.3). The two should differ only by run-to-run scatter, and they do: the mean relative change in actinic flux, taken over every element of the field (Section 4.2.2), is 0.013553% (`e1.aflux.meanrel`), and in both runs the solver's own budget of every chemical element closes to better than one part in ten thousand (keys `e1.A0.atom_loss` and `e2.A1.atom_loss`). The control is therefore the model as supplied.

Experiment 2 switched the absorber on. Figure 5.1 shows what it does. The absorber adds optical depth in and above the clouds, and the radiation field changes mainly between 300 and 420 nm and between 69 and 93 km. The largest single change in actinic flux is -4.6982×10^{12} photons per square centimetre per second per nanometre, at 81 km and 328 nm (`e2.aflux.delta_at_arg`), a fall from 7.2246×10^{13} to 6.7548×10^{13}. Over all lit elements of the field the mean relative change is 3.7078% and the largest is 87.404% (`e3fp.bnd.A1.aflux.mean_pct` and `.max_pct`).

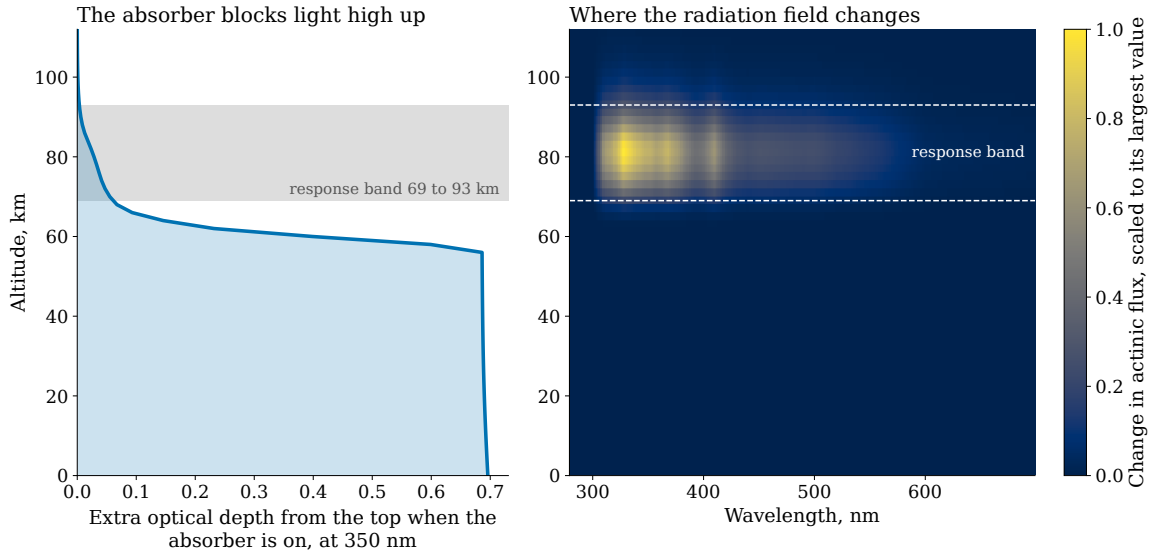


Figure 5.1: What the absorber does when it is switched on (Experiments 1 and 2). Left: the extra optical depth, counted from the top of the atmosphere, that the absorber adds at 350 nm. Right: the change in actinic flux, scaled to its largest value; it sits between 69 and 93 km and mostly between 300 and 420 nm. Sources: committed output of Experiments 1 and 2; registry keys `e6.interval.90.lo` and `.hi`.

The composition responds far less. The largest change in the SO_2 mixing ratio is 0.2243%, and in SO 1.7399%, both at 71 km (`e2.SO2.maxrel` and `e2.SO.maxrel`). Both are smaller than the reproducibility floor of Section 4.6, so they are upper bounds. A strong change in radiation that leaves the main sulfur species almost unmoved is the pattern that every later experiment had to explain, and it is why the response was next traced in amount, wavelength, reaction and altitude.

5.2 Experiment 3: is the response proportional?

If the response grows in proportion to the amount of absorber, a result found at one amount holds at any other, and later experiments can compare candidates of different strength fairly. Besides the control (amplitude 0, run A0) and the nominal run of Experiment 2 (amplitude 1, run A1), Experiment 3 added two runs, A0.5 at half and A2 at twice the nominal amplitude.

The test compares the change at half amplitude with half the change at full amplitude, element by element; a ratio of exactly 1 means proportional. Over the absorber's own wavelength range, 300 to 698 nm, the median ratio is 1.00002 for optical depth and 1.00208 for actinic flux. Outside that range the two runs differ from the control only by round-off, thousands of times smaller than the real changes inside it, so a ratio there carries no information. A second test, on the whole actinic flux field at once, gives 1.010795 at half amplitude and 0.979304 at twice the

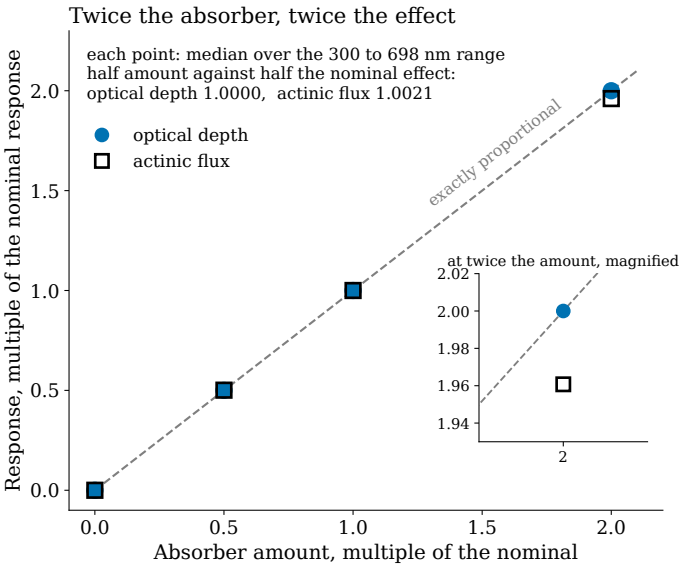


Figure 5.2: The radiative response is proportional to the amount of absorber (Experiment 3). Each point is the response at that amount as a multiple of the response at the nominal amount. Registry keys `e3.lin.tau.inband.median` and `e3.lin.aflux.inband.median`.

amplitude. Figure 5.2 shows the result.

The keys for these numbers, in order, are `e3.lin.tau.inband.median`, `e3.lin.aflux.inband.median`, `e3fp.aflux.ratio.A0p5` and `e3fp.aflux.ratio.A2p0`; the round-off comparison is `e3.lin.tau.outband.median_relsize` against `e3.lin.tau.inband.median_relsize`.

The radiative response is therefore proportional to the amount of absorber across a factor of four. The same test on the sulfur reservoirs gives ratios far from 1 at half amplitude: for SO_2 it is 1.916 (`e3fp.L2.A0p5.SO2.ratio`). This could be read as a sign that the chemistry responds non-proportionally, but that reading cannot be supported. The SO_2 and SO_3 changes are below the reproducibility floor: two runs that the radiation cannot tell apart differ in their reservoirs by about as much (Section 4.6). SO alone goes above it, at most 3.316% at half and 4.379% at twice the amplitude, inside the band at 71 km (`e3fp.A0p5.SO.maxrel_pct` and `e3fp.A2.SO.maxrel_pct`); since the floor is the smallest the scatter can be (Section 4.6), that margin is too narrow to settle the question. The radiative result stands; the chemical one is not resolved.

5.3 Experiment 4: which wavelengths?

Experiment 4 asked where in wavelength the response comes from, in two stages.

Stage 4A took the control and the nominal run (A0 and A1) and split each photolysis rate change at 81 km, the middle of the response, into the wavelength windows it came from, using VULCAN's own rule (Section 4.2.3). Different molecules draw on different parts of the spectrum. S_2O takes 93.689% of its change from 300 to 320 nm (`e4a.S2O_1.300_320.pct`); S_3 takes 49.605% from 380 to 420 nm; S_4 takes all of it from beyond 420 nm. SO_2 itself takes none of its change from these windows, because in the model it is broken apart only by light shorter than 225 nm (its photolysis cross section, `cross_J`, is zero above that). Its photolysis therefore changes only indirectly, a point Experiment 7 confirms.

Table 5.1 gives the split for ten sulfur and chlorine photolysis branches, ordered from the shortest wavelengths to the longest. S_2O and SO_3 take nearly all of their change from below 340 nm; S_3, ClS_2 and SCl take most of theirs above 380 nm; OSSO, SCl_2 and Cl_2 draw on the whole range. A candidate that absorbs at some wavelengths therefore affects the photolysis of some molecules and not others, and which molecules depends on where its absorption lies.

Table 5.1: Where in wavelength the change in each photolysis rate coefficient comes from, when the absorber is switched on (Experiment 4, Stage 4A), at 81 km, in per cent of the change. Ten sulfur and chlorine branches; keys `e4a.<branch>.<window>.pct`. A row can fall short of 100 by the small part from outside these windows.

	Window (nm)					
Branch	300 to 320	320 to 340	340 to 360	360 to 380	380 to 420	420 to 620
S_2O	93.689	6.299	0	0	0	0
SO_3	80.754	18.795	0	0	0	0
S_2Cl_2	45.016	43.215	8.574	2.872	0.319	0
Cl_2	17.513	40.010	25.976	11.616	4.207	0.679
OSSO (S_2O_2)	13.535	25.173	21.336	19.184	14.786	5.985
SCl_2	11.427	29.895	23.971	15.863	15.974	2.869
S_3	0	0	7.858	30.566	49.605	11.970
ClS_2	0	5.785	15.369	26.717	40.776	11.353
SCl	0	0.669	5.392	10.102	31.024	52.812
S_4	0	0	0	0	0	100

Stage 4B tested this with six new runs, B1 to B6, each with the absorber spectrum kept only inside one window: 300 to 320, 320 to 340, 340 to 360, 360 to 380, 380 to 420 and 420 to 698 nm, in that order. Figure 5.3 shows that each run changes the radiation field only where its absorber absorbs; outside its window the change is too small to see at this scale. The response is *spectrally local*: what the absorber does at one wavelength does not depend on what it does at another. This is what makes a test of spectral shape meaningful in Experiment 12. The six runs also share one response altitude, which Section 4.5 used to show that the response band belongs to the atmosphere, not to the spectrum.

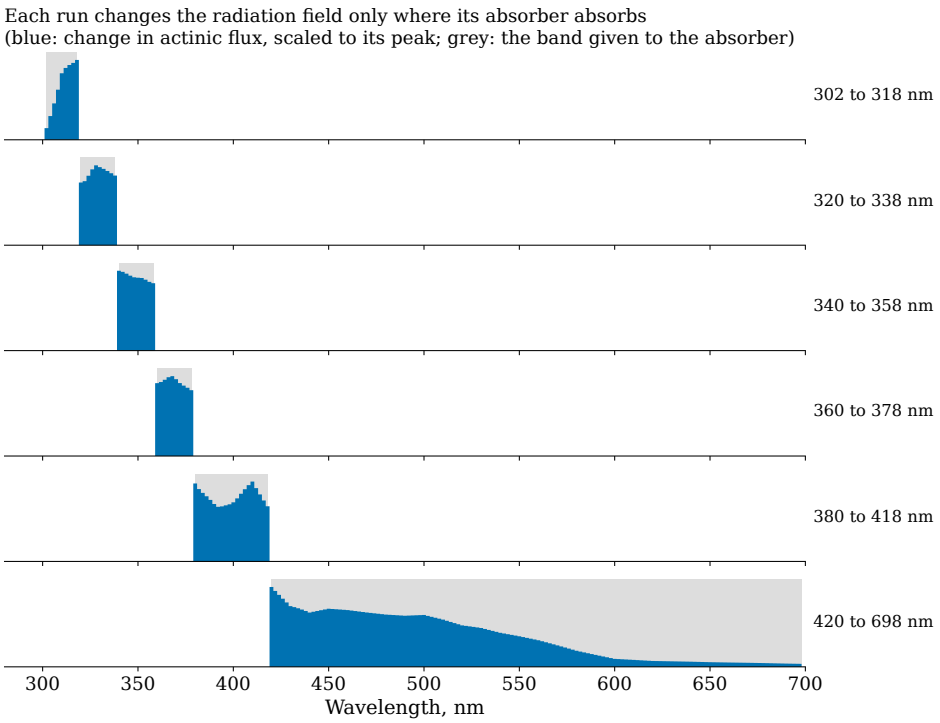


Figure 5.3: Each band-limited run changes the radiation field only inside its own window (Experiment 4, Stage 4B). Blue: change in actinic flux, scaled to its peak. Grey: the wavelengths given to the absorber in that run. Registry keys `e4b.<run>.support_lo` and `.support_hi`.

5.4 Experiment 5: which reactions?

VULCAN writes its chemistry as program code: the rate of change of each species is a long sum of signed terms, one for each reaction that makes or destroys it. For SO_2 the sum has 63 terms from 58 distinct reactions (`e5.SO2.nterms` and `.nunique`). Experiment 5 read these terms out of the generated file `chem_funs.py` and evaluated each one in both runs. Adding the terms back together gives the model's own rate of change exactly, with zero error, for all five sulfur species examined and in both runs (keys ending `extraction_error`). The attribution is therefore the model's own arithmetic.

Figure 5.4 ranks the reactions by how much their contribution to SO_2 changes. The largest change is in R593, COS + SO_3 + SO_3 → 3 SO_2 + CO, at 1.8644×10^5 molecules per cubic centimetre per second at 35 km (`e5.SO2.R593.maxdelta`). But that reaction runs at 5.7568×10^8 there (`e5.SO2.R593.bg_A0`), so it changes by only 0.032385%. The oxidation O + SO + M → SO_2 + M (R663) changes by far less in absolute terms but by 1.3809% of its own rate, at 71 km (`e5.SO2.R663.frac_pct`). The absorber reaches the sulfur chemistry through several coupled reactions, not one, and the largest absolute change does not mark the most affected reaction.

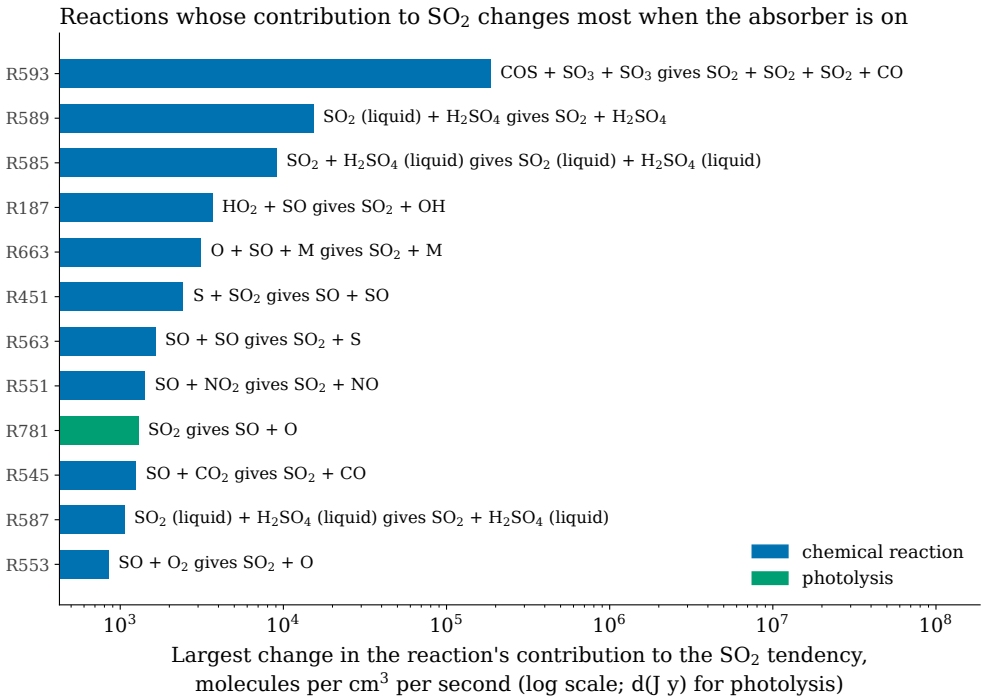


Figure 5.4: The reactions whose contribution to SO_2 changes most when the absorber is switched on (Experiment 5), on a logarithmic scale. For the photolysis R781 the quantity is the change in rate, $\Delta(Jn)$, written d(J y) on the axis because the model calls the number density y. Source: registry keys `e5.SO2.<reaction>.maxdelta`; reaction labels from `thermo/Networks/Nominal.txt`.

Changes of this size in reaction rates that depend on abundances also sit below the reproducibility floor, so the experiment shows *where* the response enters the chemistry more reliably than *how much*.

5.5 Experiment 6: which altitudes?

Experiment 6 located the response in altitude. Its radiative part defines the response band, 69 to 93 km, set out in Section 4.5; Figure 5.5 shows the profile. The experiment also located the reactions, using the change in rate, $\Delta(Jn)$ (Section 4.2.5). The photolysis reactions that the absorber acts on directly sit in the same band. Of the change in SO_2 photolysis (R781), 99.128% falls between 69 and 93 km, with its peak at 71 km (`e6.R781.pct_in_band` and `.peak_z`); for SO_3 photolysis (R787) the share is 99.992%, peaking at 73 km. Some large changes in other reactions lie much lower: R593 peaks at 35 km and R589 at 53 km, with almost nothing in the band. These are knock-on effects carried through the chemistry and by mixing, not direct effects of the absorber on sunlight.

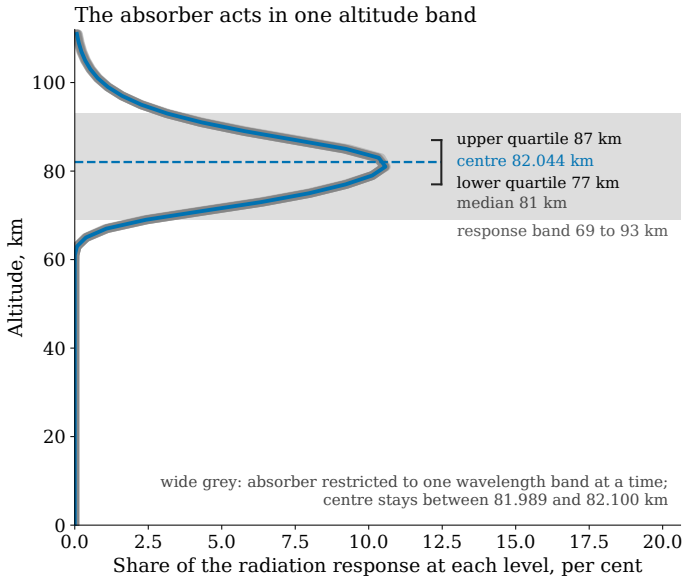


Figure 5.5: The share of the radiative response at each altitude (Experiment 6). Blue: the full absorber. Wide grey lines: the six band-limited runs of Experiment 4. Registry keys `e6.A1.meanz`, `e6.A1.q25`, `.q50`, `.q75`, `e6.interval.90.lo` and `.hi`, `e6.B5.meanz` and `e6.B6.meanz`.

5.6 Experiment 7: does it matter where the absorber sits?

Experiment 7 moved the whole absorber profile up in one run (Experiment 7A) and down in another (7B), keeping the spectrum, the amplitude and the total number of particles fixed at 4.0776×10^8 per square centimetre (`e7.nominal.column`). Only the height changed. Its effect is measured by ΔF of Equation (4.5), the size of the change in the whole actinic flux field as a percentage of the field with the absorber in its usual place.

The nominal profile is centred at 63.419 km (`e7.nominal.centroid`). Moving it down by 18 km gives a centre of 45.419 km, a shift of exactly -18.0000 km. Moving it up by 18 km gives 81.263 km, a shift of only $+17.8439$ km (registry key `e7.7a.centroid_shift`), because the top of the profile runs off the top of the model and the rest is rescaled to keep the total fixed. The two moves are therefore not mirror images, and the shift must not be written as ± 17.844 km.

Figure 5.6 shows the result. Moved up, the absorber changes the radiation field by 2.2498% (`e7a.L2pct`); moved down, by 0.3067% (`e7b.L2pct`). The same amount of the same absorber has 7.34 times more effect when it sits higher (`e7.ratio_7A_over_7B`). The responding layer itself barely moves: the response is centred at 82.435 km in the upward case and 82.355 km in the downward case

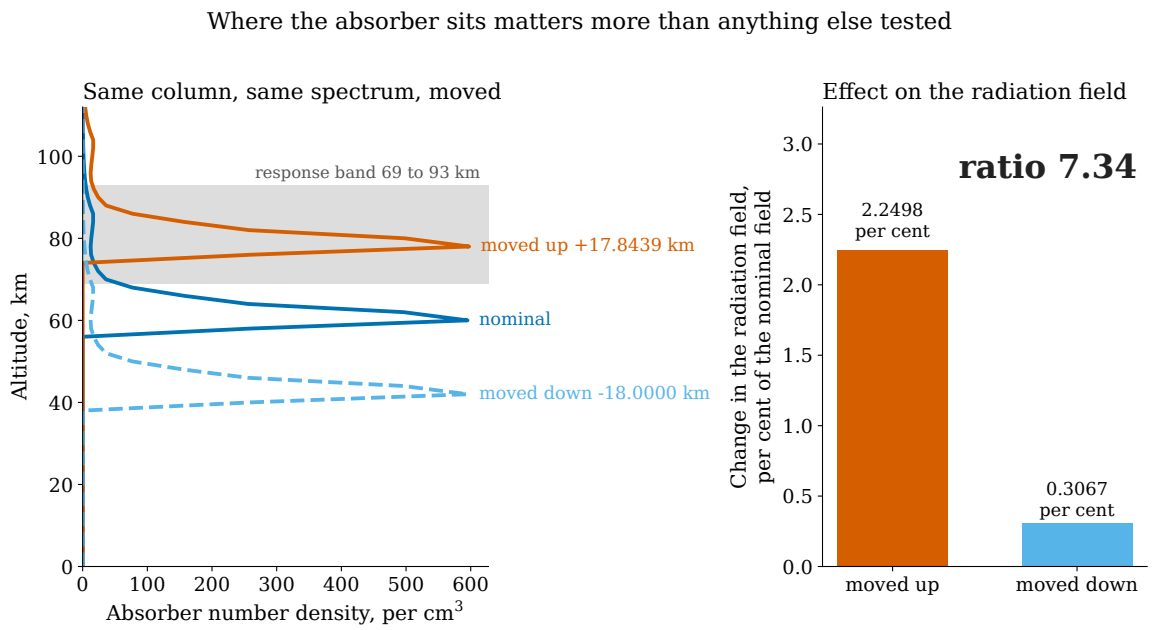


Figure 5.6: The same absorber, moved up or down, changes the radiation field by very different amounts (Experiment 7). Left: the three absorber profiles, with the same total. Right: the change in the radiation field as a percentage of the nominal field. Registry keys `e7.7a.centroid_shift`, `e7.7b.centroid_shift`, `e7a.L2pct`, `e7b.L2pct` and `e7.ratio_7A_over_7B`.

(`e7a.meanalt` and `e7b.meanalt`), close to the 82.044 km of Experiment 6. An absorber high up overlaps that layer; the same absorber low down mostly shades light that has already been weakened.

The chemistry confirms the picture. Three reactions whose rates are set by the local light, R663, R551 ($SO + NO_2 \rightarrow SO_2 + NO$) and R665 ($O + SO_2 + M \rightarrow SO_3 + M$), change in opposite directions when the absorber moves up and when it moves down (keys `e7.7a` and `e7.7b` ending `.col`). Three others, R187, R563 and R781, change in the same direction both ways; why is not yet known, and it is recorded as an open question, not a result. Table 5.2 gives the numbers: for each reaction, the change in its contribution to the SO_2 budget, added over the response band with the thickness in centimetres (Section 4.2.4). The upward move changes each of them several times more than the downward move, as it changes the radiation field several times more.

The upward move changes SO and SO_3 in the band by 8.404% and 4.718% (`res.e7a.SO.inband` and `res.e7a.SO3.inband`), the largest changes in the sulfur reservoirs in these seven experiments and well above the reproducibility floor; SO crosses the floor narrowly in Experiment 3 as well, and changes by up to about 3% in some Experiment 4B runs (`e4b.B2.SO.maxrel`). The SO_2 column falls slightly in both cases, by 0.00048557% and 0.0016987% (`res.e7a.SO2.col`

Table 5.2: Change in each reaction's contribution to the SO_2 budget, added over the response band, in molecules per square centimetre per second (Experiment 7). A positive value means the reaction adds more SO_2 or removes less. Keys `e7.7a.SO2.<reaction>.col` and `e7.7b.SO2.<reaction>.col`.

Reaction	Equation	Moved up (7A)	Moved down (7B)
R663	O + SO + M $\rightarrow$ SO_2 + M	-5.6390×10^9	$+6.6392 \times 10^8$
R551	SO + NO_2 $\rightarrow$ SO_2 + NO	$+6.7343 \times 10^9$	-6.7855×10^8
R665	O + SO_2 + M $\rightarrow$ SO_3 + M	$+3.1466 \times 10^8$	-6.1425×10^7
R187	HO_2 + SO $\rightarrow$ SO_2 + OH	-3.8755×10^9	-5.2023×10^8
R563	SO + SO $\rightarrow$ SO_2 + S	$+1.3783 \times 10^9$	$+3.1740 \times 10^8$
R781	SO_2 + $h\nu$ $\rightarrow$ SO + O	$+8.2352 \times 10^8$	$+2.2860 \times 10^8$

and `res.e7b.SO2.col`), so bulk SO_2 does not follow the absorber's height in any simple way.

5.7 What Experiments 1 to 7 establish

- The absorber term works and the control is exact. Switched on, the absorber changes the radiation field strongly and the main sulfur species very little.
- The radiative response is proportional to the absorber's amount, and it is local in wavelength.
- The response lives in one band of altitude, 69 to 93 km, whatever part of the spectrum the absorber occupies. It reaches the chemistry through several coupled reactions.
- Where the absorber sits matters: the same column, moved up instead of down, has 7.34 times the effect. Placement is a first-order control, and so a candidate's height has to be tested, not assumed. Chapter 6 tests it.

Chapter 6

Experiments 8 to 14

Chapter 5 ended with one firm result: the same absorber has 7.34 times more effect when it sits higher. This chapter compares that with spectral shape (Experiment 8), lets the model's own chemistry place a candidate (Experiment 9), measures how far two runs that should agree can differ (Experiment 10), and then tests published spectra, every species the model can represent, iron minerals and ferric chloride (Experiments 11 to 14). A change in the radiation field is measured as in Section 5.6, as a percentage of the reference field.

6.1 Experiment 8: spectral shape at a fixed placement

Question and design. If the amount and height of the absorber are held fixed, how much does the shape of its spectrum matter? Three runs put the spectra of ClS_2 (Experiment 8A), OSSO, that is S_2O_2 (8B), and S_3 (8C) in place of the empirical spectrum, each scaled so that its absorption added over 300 to 698 nm equals the empirical one, on the unchanged profile. The optical depth of the absorber above 69 km is then 8.2458 for every candidate (`spec_8A.tau69`) against 8.1928 for the empirical spectrum (`spec_emp.tau69`), only 0.6461% apart (`tau.spec_change_pct`). The runs differ only in where along the wavelength axis the absorption sits.

Result. The radiation field changes by 0.4342% for ClS_2, 0.1084% for OSSO and 0.5369% for S_3 (`e8a.L2pct`, `e8b.L2pct`, `e8c.L2pct`; Figure 6.1). The largest is 4.95 times the smallest, so spectral shape is a real, independent control. But moving the absorber changed the effect 7.34-fold, and the largest spectral effect is well below the 2.2498% of the upward move. *Placement matters more than spectral shape, and spectral shape is not negligible.* This comparison is the quantitative core of Phase 1. One caution: the responses are in the reverse order of each spectrum's correlation with the empirical one, 0.9394 for OSSO, 0.3419 for ClS_2 and 0.2253 for S_3 (`corr.8B`, `corr.8A`, `corr.8C`). Since the empirical spectrum was fitted to the observed brightness of Venus, the ranking measures resemblance to that fit, not correctness.

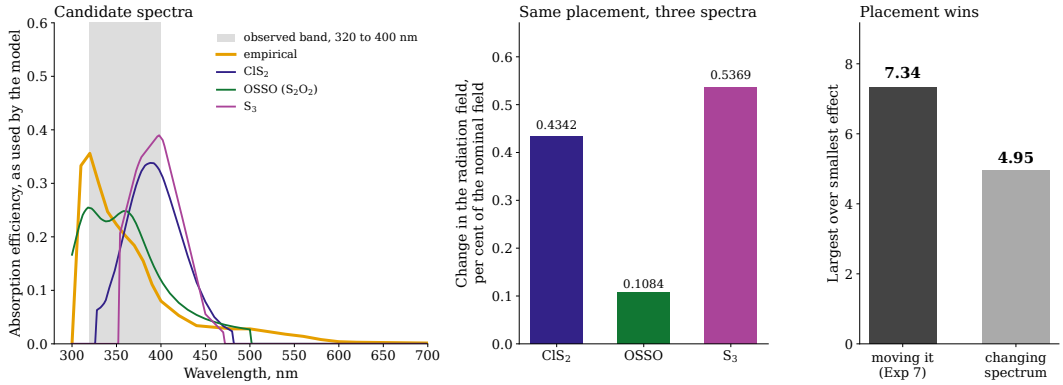


Figure 6.1: Spectral shape against placement (Experiment 8). Left: the spectra as the model uses them. Middle: change in the radiation field for each candidate at the same placement. Right: largest over smallest effect, for moving the absorber (Experiment 7) and for changing its spectrum. Keys `e8a.L2pct`, `e8b.L2pct`, `e8c.L2pct`, `e7a.L2pct` and `e7b.L2pct`.

The amount check. A script that only reads the saved baseline divided the required optical depth τ_{req} (Section 3.6) by the optical depth that each of twelve sulfur and chlorine species gives with its column above 58 km, at the wavelength where its absorption peaks within 300 to 620 nm. The result, the *shortfall factor*, is how many times too little of the species there is. ClS_2 comes closest, short by 121.914 at 388 nm (`scr.ClS2.short`), about two orders of magnitude; next is S_2O at 6070, and the rest are short by 4.4×10^4 (`scr.S2.short`) to 4.2×10^8 (`scr.SCl.short`). OSSO is short by 4.2×10^6 (`scr.S2O2.short`). Egan et al. found OSSO about three orders of magnitude too low by a different route [6]. The two agree in direction and differ in size because they measure against different things, and theirs uses SO_2 above the clouds kept deliberately high to give OSSO every chance [6]. OSSO is the only candidate with an independent published verdict, so this agreement is the main outside check on the screen's arithmetic. ClS_2, the one species close in amount, went forward to Experiment 9. Table 6.1 lists all twelve.

6.2 Experiment 9: letting the chemistry place the candidate

Question and design. Experiments 7 and 8 used the empirical profile, a height chosen by the modeller. What happens when a candidate sits where the model's own chemistry puts it? The ClS_2 spectrum was kept, and the absorber profile was given the shape of the model's own steady-state ClS_2 profile, rescaled so that the sum of the absorber over all levels stayed the same. The expected outcome was written down before the run: if the absorber left the response band, the run should reproduce the difference between the runs with and without the absorber

Table 6.1: The amount check of Experiment 8: each species' column above 58 km, at the wavelength where its absorption peaks within 300 to 620 nm, against the required optical depth. Keys `scr.<species>.lam` and `scr.<species>.short`.

Species	Peak wavelength (nm)	Shortfall factor
ClS_2	388	1.219×10^2
S_2O	300	6.070×10^3
S_2	300	4.415×10^4
S_3	398	7.486×10^4
S_4	500	9.962×10^4
ClO	300	6.754×10^5
SCl_2	338	8.049×10^5
S_2Cl_2	300	1.092×10^6
Cl_2	328	1.604×10^6
OSSO	318	4.207×10^6
SO_3	300	8.626×10^6
SCl	470	4.205×10^8

(Section 5.5), with the opposite sign.

Where the chemistry puts it. The empirical absorber is a layer at the cloud top, centred at 63.419 km (`pnom.centroid`), with 9.1940% of its column inside the response band (`pnom.inband`). The model's ClS_2 is centred at 13.656 km (`p9.centroid`), 51.521 km lower (`e9.centroid_shift`), with 91.80% of it below 30 km (`p9.below30`) and 0.0000% inside the band (`p9.inband`; Figure 6.2). The registry shows where the chemistry puts it, not why; how fast ClS_2 is made and destroyed at each height is the quantity Phase 2 measures first (Chapter 9, WP5). The optical depth above 69 km falls from 8.1928 to 4.0153×10^{-8} (`spec_9.tau69`), a factor of 2.0×10^8 (`tau.reduction`). Keeping the level sum fixed left the column (Section 4.2.4) 12.88% lower (`p9.col_deficit_pct`), because the new profile puts that share of its level sum in the model's lowest layer, which is only 2 cm thick; beside a factor of 2.0×10^8 this cannot matter.

Result. The prediction held. The peak change in actinic flux has the size and place of the no-absorber control, at 81 km and 328 nm, with the opposite sign. Added cell by cell, the two change fields leave 0.001487% of the control (`id.l2_ratio_pct`), and over the 3425 cells carrying more than 1% of the peak their ratio has mean 0.99994 and standard deviation 9.4×10^{-5} (`id.ratio_mean`, `id.ratio_sd`).

This is the central result of Phase 1. Placed where the model's own chemistry puts it, ClS_2 is radiatively the same as no absorber at all in the band where the absorber acts. It is in the calculation at full strength, with a spectrum in the right range, but in the wrong place. A candidate can come close in amount and have a suitable spectrum and still be invisible, because the chemistry places it below the altitudes where the radiation responds. This failure is separate from its shortfall in amount.

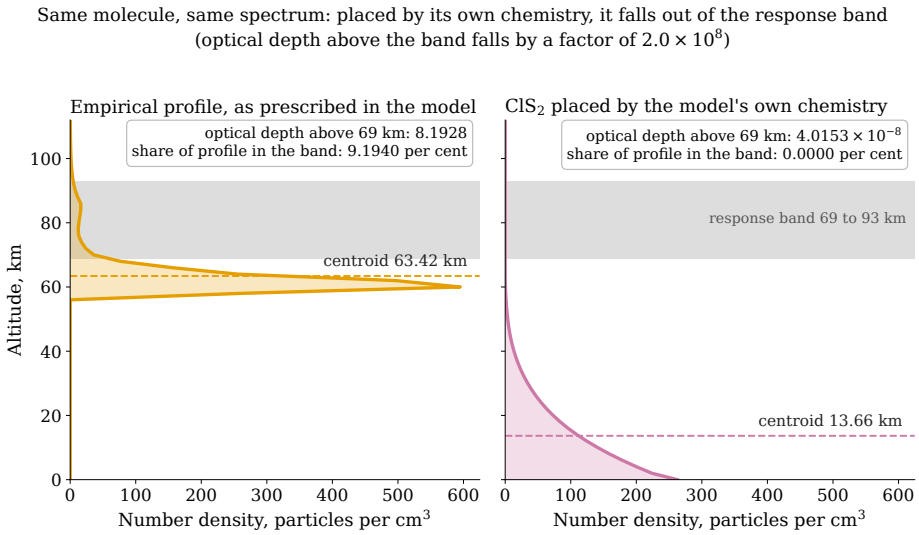


Figure 6.2: The same molecule and spectrum, placed two ways (Experiment 9). Left: the empirical profile. Right: the profile the model's own chemistry gives ClS_2. The grey band is the response band. Keys `pnom.centroid`, `p9.centroid`, `spec_emp.tau69`, `spec_9.tau69`, `pnom.inband`, `p9.inband` and `tau.reduction`.

The two photolysis conventions of Section 4.2.5 give different pictures here, and both are correct. Nearly every rate coefficient rises a little, the largest ΔJ being 3.168% for S_2O_2 at 81 km (`dJ.e9.S2O2`); those of SO_2 and SO fall very slightly (`dJ.e9.SO2`, `dJ.e9.SO`). But the rate $\Delta(Jn)$ of S_2Cl_2 and SCl_2 photolysis falls by 21.751% and 19.630% at 69 km (`dJy.e9.S2Cl2`, `dJy.e9.SCl2`). Since the coefficients rise, these falls come from smaller amounts of the two molecules, and those amounts are abundances, which the floor of Section 4.6 limits; no cause is claimed for them. The largest change among the sulfur reservoirs in the band, SO by 0.7384% (`res.e9.SO.inband`), is below that floor, so no abundance result is claimed.

6.3 Experiment 10: a replicate, two floors and a retraction

Experiment 10 put the S_3 spectrum on the same ClS_2 profile, to test whether a pattern seen in Experiment 9 held for a third combination. The test was void: both spectra lie wholly inside the range used for scaling and were scaled to the same total on the same profile, so the optical depth above 69 km is identical to every digit kept (`tau.9_minus_10` is zero), and the difference between the two actinic flux fields is only 0.0017% of the change that Experiment 9 makes (`e10.fielddiff_pct`; Section 4.4). Experiment 10 is a numerical replicate of Experiment 9, and that accident gave three results.

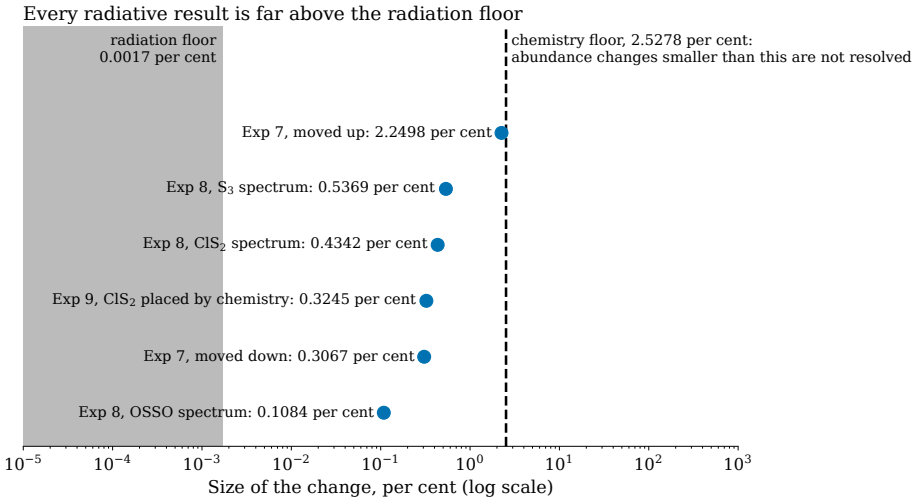


Figure 6.3: The two floors (Experiments 9 and 10). Every radiative result lies far above the radiation floor of 0.0017% (`e10.fielddiff_pct`), which is a share of the change Experiment 9 makes and so smaller still as a share of the field; the chemistry floor of 2.5278% (`res.e10.SO.inband`) applies to abundances only.

Two floors, not one. Since the radiation fields agree, any difference in the chemistry is numerical. SO in the band differs from the baseline by 0.7384% in one run and 2.5278% in the other (`res.e10.SO.inband`): this is the chemistry floor of Section 4.6. Figure 6.3 sets it beside the radiation floor of 0.0017%. The two numbers are not on the same footing. Every radiative result, from 0.1084% for OSSO to 2.2498% for the upward move, is a share of the field (Equation (4.5)); the radiation floor is a share of the change Experiment 9 makes, and that change is itself only 0.3245% of the field (`e9.L2pct`). Two runs that should agree therefore differ by far less than any result, so every radiative comparison stands; the chemistry floor limits only claims about abundances. Only two abundance changes in the project exceed it clearly: those of SO and SO_3 when the absorber is moved up (Section 5.6); SO also crosses it narrowly in Experiments 3 and 4B.

Spectral shape matters only where the absorber overlaps the responding layer. On the empirical profile, S_3 gives 1.2365 times the effect of ClS_2; on the ClS_2 profile the same ratio is 1.000003. Out of the band, the spectrum no longer matters (keys `twoby2.nominal_ratio` and `twoby2.cls2_ratio`).

A retraction. In Experiment 9 the change in each of nine photolysis rate coefficients was the change in Experiment 7B times one constant, 1.0474 with standard deviation 0.00056 (`chr.mean`, `chr.sd`); the nine ratios are ClS_2 1.0473, H_2SO_4 1.0466, S_2Cl_2 1.0482, S_2O 1.0480, OSSO 1.0477, S_3 1.0472, S_4 1.0466, SCl 1.0469, SCl_2 1.0477 (keys `chr.<species>`). This had been read as a "fingerprint" of absorbers below the cloud top. It is not: the ratio of the actinic flux changes them-

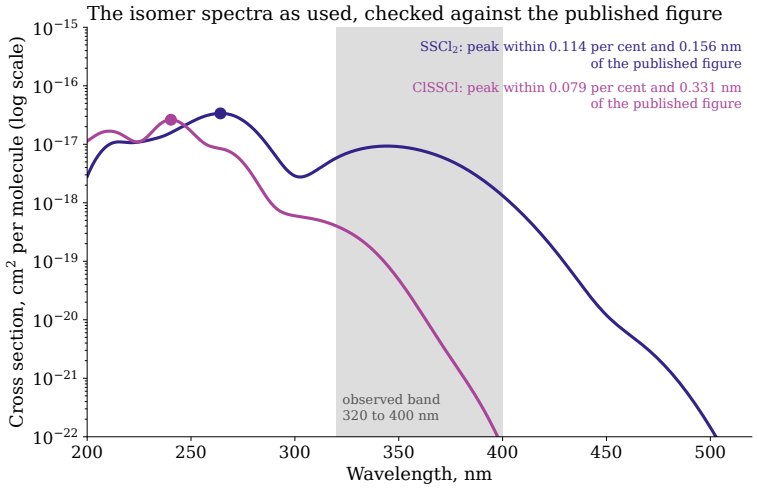


Figure 6.4: The isomer cross sections as used, checked against the published figure (Experiment 11). Keys `iso.SSCl2.val_pct`, `iso.SSCl2.val_nm`, `iso.ClSSCl.val_pct` and `iso.ClSSCl.val_nm`.

selves at 81 km, the peak response height, added over all wavelengths, is 1.0473 (`chr.flux_ratio`). The constant is linear response; when two small disturbances change the light in the same place, every photolysis rate follows the light. For Experiment 7A, where the optical depth above 69 km is 10.1975 times the nominal (`tau.7A_over_A1`), the ratios for eight of the species scatter widely (mean -14.12, spread 12.43; `chr8_7a_7b.mean` and `.spread`), which locates the edge of the linear regime. The fingerprint reading is withdrawn in commit `5a7e2ac`.

6.4 Experiment 11: computed cross sections of two isomers

Question and design. Trabelsi and Francisco computed the cross sections of two chlorine and sulfur isomers, ClSSCl and $SSCl_2$, and offer them as model inputs [17]; the model tracks only one lumped species, S_2Cl_2. The computed data, obtained from the authors and checked against their published figure, were run as absorber spectra under the rules of Experiment 8, $SSCl_2$ in Experiment 11A and ClSSCl in 11B.

Result. Each peak matches the published figure to 0.114% in height and 0.156 nm in position for $SSCl_2$, and 0.079% and 0.331 nm for ClSSCl (Figure 6.4). $SSCl_2$ puts 85.49% of its absorption between 320 and 400 nm, more than any other spectrum tested and more than the empirical 61.58%; its correlation with the empirical spectrum, 0.88983, is second to OSSO's (keys `share320400.11A`, `share320400.emp` and `corr.11A`). It is therefore the best match by share but only the second by

correlation. On the empirical profile the field changes by 0.19465% for $SSCl_2$ and 0.30144% for ClSSCl (`e11a.L2pct`, `e11b.L2pct`). Across all five spectra the responses fall in exactly the reverse order of their correlations (rank correlation -1, `five.spearman`), with the caution of Experiment 8.

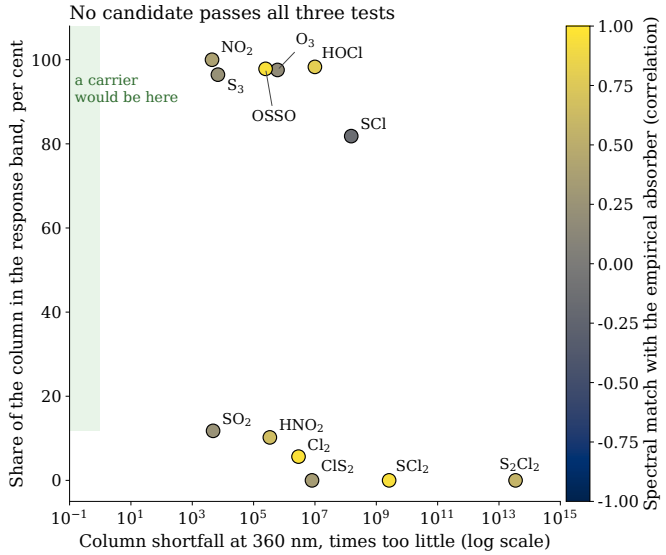
Neither isomer can be made in the amount needed: at its absorption peak within 300 to 620 nm $SSCl_2$ is short by 4.266×10^5 and ClSSCl by 6.668×10^6 (`iso.SSCl2.short`, `iso.ClSSCl.short`). ClSSCl absorbs most strongly near 240 nm, below the start of the absorber's grid at 300 nm, so the model sees only its tail, scaled up to a peak efficiency of 0.9467 against the empirical 0.356 (`spec_11B.peakQ`); its response describes that tail, not the molecule. Within the literature reviewed in Chapter 2, no earlier study has run computed chlorine and sulfur cross sections in a Venus photochemistry model; this is the extent of the review, not a certainty.

6.5 Experiment 12: all three tests on every species

Question and design. Which species the model can represent pass amount, placement and spectrum together? A read-only script scored every species in the network with a cross section between 300 and 620 nm: 23 of the 62 tracked (`e12.n_screened`, `e12.n_tracked`). Nineteen have no cross section in that range, 18 have none at all, and the two cloud particle tracers are excluded; a species absent from the network, such as ferric chloride, cannot be screened. Two choices, learned from Experiment 9, differ from the Experiment 8 check. A species is credited only with its column *above 69 km*, and it is compared with the absorber *at each wavelength*, not at its own peak, which would favour species whose absorption falls steeply from below 300 nm. SO_2 shows why: at its own peak it looks only 2.402 times short above 58 km (`e12.SO2.shortfall58`), but at 360 nm above 69 km it is short by 4768.6 (`e12.SO2.short.360`). ClS_2 goes from 121.914 to 8.0×10^6 (`e12.ClS2.short.360`), because almost none of it is above 69 km.

The screen itself sets no pass marks. The marks used here (Figure 1.3) are set so that the empirical absorber itself passes every test: a shortfall at 360 nm of at most 1, which the absorber meets exactly; at least the absorber's own 11.92% of the column above 58 km inside the band (`e12.gate.emp_inband_pct`); and more than half of the absorption inside 320 to 400 nm, with a correlation above 0.5.

Result. Table 6.2 gives every species. Sixteen species pass on placement and five on spectrum. Two pass both, OSSO and HOCl, yet in amount they are short by 2.4563×10^5 and 9.9572×10^6. None passes on amount, so none passes



Best eight on each test, best first

smallest shortfall	most column in the band	best spectral match
NO_2	NO_3	Cl_2
SO_2	NO_2	OSSO
S_3	$ClCO_3$	SCl_2
OSSO	SO_3	HOCl
HNO_2	ClO	HNO_2
O_3	S_2	S_2Cl_2
Cl_2	HOCl	H_2O_2
ClS_2	OSSO	NO_2

Each test picks a different winner: the rankings disagree.

no absorption at this wavelength, so not placed: $ClCO_3$, ClO, H_2O_2, H_2SO_4, HNO_3, NO_3, S_2, S_2O, S_4, SO_2Cl_2, SO_3

Figure 6.5: The three-axis screen (Experiment 12): shortfall at 360 nm (horizontal), share of the column in the response band (vertical) and spectral correlation (colour). A carrier would sit in the shaded corner, with no shortfall and at least the absorber's own share of its column in the band, 11.92% (`e12.gate.emp_inband_pct`); species that do not absorb at 360 nm are listed below the axis. The right panel lists the best eight species on each test. Keys `e12.X.short.360`, `e12.X.frac_in_band_pct` and `e12.X.corr`, where X is the species.

all three (Figure 6.5; keys `e12.gate.n_placement`, `e12.gate.n_spectrum`, `e12.gate.n_all`, `e12.S2O2.short.360` and `e12.HOCl.short.360`). The tests also disagree about which species is "best". NO_2 is the least short, by a factor of 4402.887 (key `e12.best_shortfall`), and has 99.998% of its column in the band, but peaks at 430 nm with a correlation of only 0.3862. HNO_2 has the most absorption inside 320 to 400 nm, 98.031%, but only 10.216% of its column in the band. Cl_2 correlates best, at 0.9461, with 5.647% in the band. No single "most likely" ranking exists.

Amount is a gate, not a tie-breaker: the shortfalls run from thousands to about 10^{13}, while placement and spectrum were each worth less than a factor of ten.

6.6 Experiment 13: iron and sulfur minerals

Question and design. Jiang et al. proposed rhomboclase and acid ferric sulfate from their spectral shape [9]. Spacek et al. later derived how strongly the droplet liquid must absorb: a decadic coefficient of 1278 per centimetre at 375 nm [15]. Nobody had put the two together. Using the mass extinction coefficients measured by Jiang et al. and the straight-line scaling of Spacek et al., the mass of mineral needed per

Table 6.2: The three-axis screen for all 23 species (Experiment 12), ordered by shortfall. Shortfall: times too little at 360 nm above 69 km ("none at 360 nm": the species does not absorb there). In band: per cent of the column above 58 km that lies in the response band. Share: per cent of the absorption between 300 and 698 nm that lies in 320 to 400 nm. Pass marks: shortfall at most 1; in band at least 11.92 per cent; share above 50 per cent with correlation above 0.5. Keys `e12.<species>.short.360`, `.frac_in_band_pct`, `.share_320_400_pct`, `.corr` and `e12.gate.*`.

Species	Shortfall	In band (%)	Share (%)	Correlation	Passes on
NO_2	4.403×10^{3}	99.998	36.140	0.386	placement
SO_2	4.769×10^{3}	11.772	4.272	0.239	none
S_3	6.859×10^{3}	96.447	56.926	0.225	placement
OSSO	2.456×10^{5}	97.835	64.275	0.939	placement, spectrum
HNO_2	3.384×10^{5}	10.216	98.031	0.646	spectrum
O_3	6.012×10^{5}	97.570	5.606	0.169	placement
Cl_2	2.910×10^{6}	5.647	73.719	0.946	spectrum
ClS_2	8.010×10^{6}	1.3×10^{-4}	60.673	0.342	none
HOCl	9.957×10^{6}	98.295	52.967	0.796	placement, spectrum
SCl	1.523×10^{8}	81.836	14.474	−0.156	placement
SCl_2	2.618×10^{9}	2.3×10^{-3}	71.043	0.936	spectrum
S_2Cl_2	3.469×10^{13}	5.0×10^{-6}	32.933	0.570	none
$ClCO_3$	none at 360 nm	99.981	0	0.225	placement
ClO	none at 360 nm	99.332	4.121	0.159	placement
H_2O_2	none at 360 nm	58.857	27.559	0.484	placement
H_2SO_4	none at 360 nm	0.969	0	−0.218	none
HNO_3	none at 360 nm	34.902	6.619	0.215	placement
NO_3	none at 360 nm	100.000	0	−0.519	placement
S_2	none at 360 nm	98.741	0.167	0.253	placement
S_2O	none at 360 nm	50.181	4.159	0.239	placement
S_4	none at 360 nm	97.261	0	−0.304	placement
SO_2Cl_2	none at 360 nm	35.317	0	−0.039	placement
SO_3	none at 360 nm	99.847	10.286	0.314	placement

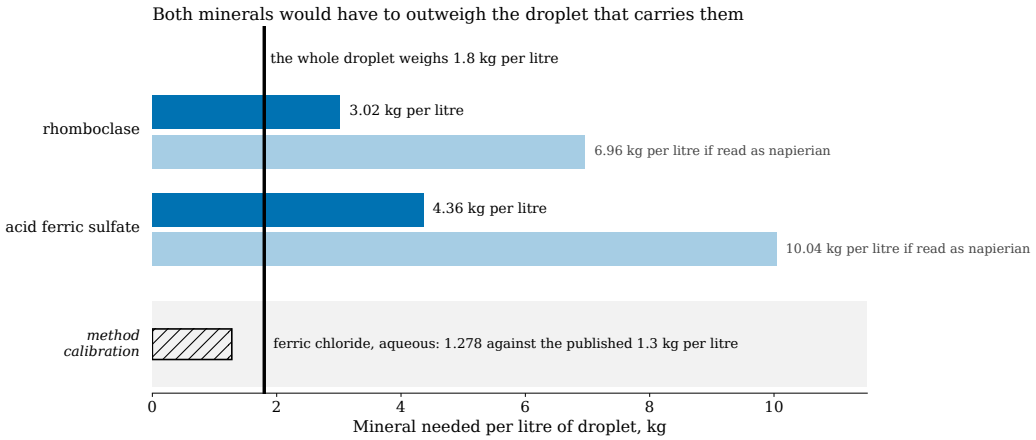


Figure 6.6: Mineral needed per litre of droplet to meet the requirement at 375 nm (Experiment 13); dark bars read the coefficients as decadic, light bars as natural (Napierian), the vertical line is the mass of a litre of droplet and the hatched bar is the calibration. Keys `e13.rhomboclase.*`, `e13.acid_ferric_sulfate.*` and `e13.fecl3.*`.

litre of droplet is the requirement divided by the coefficient:

$$m = \frac{a_{\mathrm{req}}}{\varepsilon}, \qquad s = 100\,\frac{m}{\rho}, \tag{6.1}$$

where $a_{\mathrm{req}} = 1278$ per centimetre is the required decadic coefficient at 375 nm (`e13.A_required`), ε is the material's mass absorption coefficient at 375 nm in square centimetres per gram, m is the mass needed in grams per cubic centimetre, which is the same number in kilograms per litre, and s is that mass as a share of the 1.8 kg that a litre of droplet liquid weighs, the droplet density used by Dai et al. [3] (`e13.rho_solution`). For rhomboclase ε is 422.797 and for acid ferric sulfate 293.144 (`e13.X.eps_375`); for ferric chloride in water, 1 g per litre gives about 1 per centimetre, so ε is 1000. The script first reproduces the ferric chloride figure of Spacek et al. from their stated input, 1.278 against their 1.3 kg per litre (`e13.fecl3.loading_kg_per_L`); that line is a check on the method, never a result about ferric chloride.

Result. Rhomboclase would need 3.02 kg per litre, 167.93% of the droplet's own mass, and acid ferric sulfate would need 4.36 kg per litre, 242.20% (keys `e13.X.pct_of_droplet`, where X is the mineral), against a droplet density of 1.8 kg per litre (Figure 6.6). At 375 nm, even a droplet made entirely of either mineral could not absorb enough. Reading the coefficients as natural rather than decadic raises the need to 6.96 and 10.04 kg per litre. At each mineral's own absorption peak the need is 156.43% and 95.07% of the droplet mass (`e13.X.pct_at_peak`): rhomboclase still cannot reach it, and acid ferric sulfate could only as a droplet almost wholly made of the mineral. The verdict is also conservative: the coefficients

were measured through a suspension and include scattering, and removing it can only raise the mass needed.

6.7 Experiment 14: ferric chloride measured in acid

Experiment 14 was done after Phase 1 closed; it is reported because it changes how a published result should be read.

Question and design. Spacek et al. find the absorbance of ferric chloride too low, because it would need about 1.3 kg per litre [15]; the method of Experiment 13 gives 1.278 kg per litre from the same input, which with the droplet density is 71.0% of the droplet mass (`e14.aqueous_pct_of_droplet`). Their input is an absorbance measured in water. Egan et al. note that the solvent has a significant effect on the spectrum of ferric chloride, and derived its molar absorptivity in hydrochloric acid at five strengths, from 1.54 to 12.2 moles per litre, labelling the curves Cl A to Cl E; they use these curves as the ferric chloride components when they fit their mixed sulfuric and hydrochloric acid samples [7]. One script read those five curves from the published figure at 600 dots per inch and writes nothing unless three checks pass. The first finds the frame of the left panel, so that every point is measured from the true axes. The second requires the tick marks to be evenly spaced to better than 1% on both axes, so that the scale has been read without stretching. The third requires every curve to fall back to zero at its long-wavelength end, so that the vertical zero is where the figure puts it. To test the checks, the zero was deliberately shifted; the third check first let the shift through, and was tightened to a tenth of a tick until it caught it. A second script reproduces the figure of Spacek et al. from their aqueous input, 1.278 against their 1.3 kg per litre, then repeats Equation (6.1) with the five hydrochloric acid values.

Result. In these solutions, ferric chloride absorbs 4.164 to 25.192 times more strongly at 375 nm than the aqueous value (`e14.ratio_lo`, `e14.ratio_hi`). The loading needed falls from 1.278 kg per litre to between 0.0507 and 0.3069, that is from 71.0% of the droplet mass to between 2.82% and 17.05% (`e14.ClA.pct_of_droplet`, `e14.ClE.pct_of_droplet`; Figure 6.7). If ferric chloride in the droplets absorbs as it does in any of these solutions, the rejection does not survive: even the weakest curve leaves it a plausible minor component. The answer is a range because the iron forms several chloride complexes whose shares depend on the chloride available, and how much chloride the droplets hold is not settled here. Table 6.3 shows the trend: the more hydrochloric acid in the solution, the more strongly ferric chloride absorbs at 375 nm and the less of it is needed.

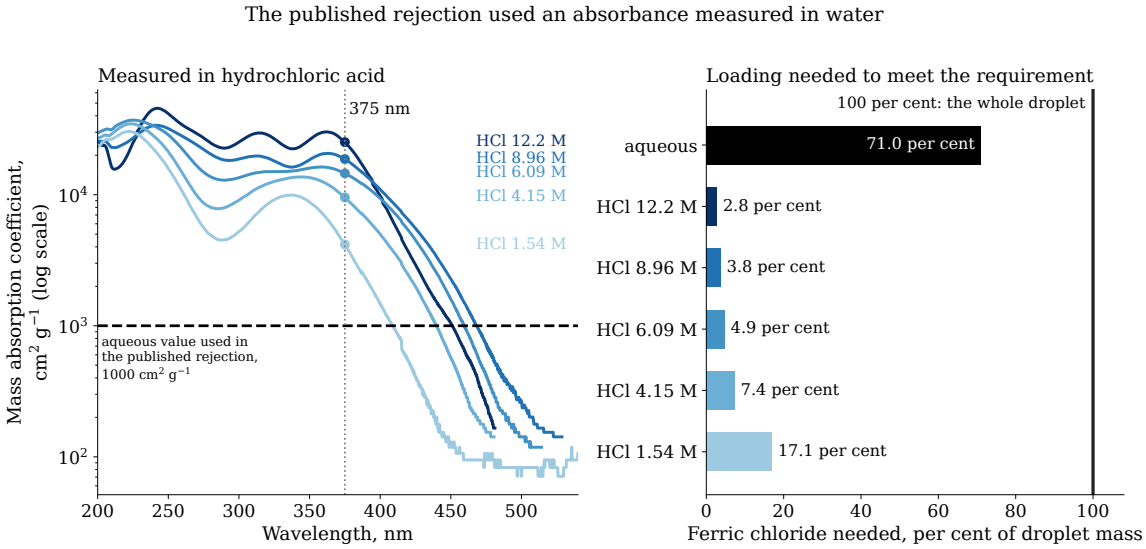


Figure 6.7: Ferric chloride in hydrochloric acid at five strengths (left, read from Egan et al. [7]) and the loading each needs as a share of the droplet mass (right), against the aqueous value of the published rejection (Experiment 14). The curves are those of their Figure 5a. Keys `e14.ClA.*` to `e14.ClE.*` and `e14.factor_to_fit_after`.

Table 6.3: The five ferric chloride curves of Egan et al. [7] at 375 nm, and the loading each needs by Equation (6.1) (Experiment 14). Molar ε in litres per mole per centimetre, mass ε in square centimetres per gram; share of the droplet's own mass. Keys `e14.<curve>.hcl_M`, `.eps_molar`, `.eps_mass`, `.loading_kg_per_L` and `.pct_of_droplet`.

Curve	HCl (mol/L)	ε, molar	ε, mass	Loading (kg/L)	Share (%)
Cl A	12.20	4086.212	25191.806	0.0507	2.82
Cl B	8.96	3036.595	18720.842	0.0683	3.79
Cl C	6.09	2362.116	14562.626	0.0878	4.88
Cl D	4.15	1546.601	9534.912	0.1340	7.45
Cl E	1.54	675.439	4164.130	0.3069	17.05

The correction also brings two 2026 papers that do not cite each other closer. The preprint of Egan et al., not peer reviewed, fits 1.0 to 1.2 wt% ferric chloride in the smallest droplets to the MESSENGER spectrum [8]. The most chloride-rich curve here needs 2.82%, a factor of 2.6 from the middle of that range (`e14.factor_to_fit_after`), against 64.5 before. The two figures still rest on different assumptions: Spacek et al. spread the absorber through two droplet sizes in a layer near the cloud top, while the preprint puts it in the smallest droplets only, which also reach higher into the haze. Spacek et al. note that either change would alter their requirement, in opposite directions [15], so the remaining factor of 2.6 is not a contradiction, but it is not yet explained.

What Experiment 14 does not claim. It corrects one input to another group's argument. It does not identify the absorber, and nothing in it measures Venus. The laboratory spectra were taken far more dilute than these loadings and are scaled in a straight line, as Spacek et al. also do. Above all, ferric chloride has not been tested on altitude, the test that decided Experiment 9, because it has no chemistry in the model and so cannot be placed by it. That test belongs to Phase 2 (Chapter 9).

6.8 What Experiments 8 to 14 establish

- Placement matters more than spectral shape, 7.34 against 4.95, and shape matters only where the absorber overlaps the responding layer.
- Placed by the model's own chemistry, the closest gas-phase candidate falls out of the response band and becomes radiatively the same as no absorber.
- Two runs that should agree give radiation fields that differ by 0.0017% of the change they make, but change the SO abundance in the band by 0.7384% and 2.5278% from the same baseline. An apparent fingerprint was linear response and has been withdrawn.
- No species the model can represent passes amount, placement and spectrum together, and the three tests disagree about which comes closest.
- At 375 nm the proposed iron and sulfur minerals cannot absorb enough even as pure solids. Ferric chloride, rejected on an aqueous measurement, becomes a plausible minor component if it absorbs in the droplets as it does in hydrochloric acid, but its altitude is untested.

Chapter 7

What Phase 1 Establishes

This chapter states what the fourteen experiments establish and within which limits. *Measured* marks a claim resting on radiative comparisons, reproducible to 0.0017% of the change itself; *bounded* marks one resting on abundances, reproducible only to 2.5278% (Experiment 10).

7.1 Established results

The absorber term works, and its effect is located (measured). The model's absorber changes the radiation field in proportion to its amount, locally in wavelength, and within one band of altitude, 69 to 93 km (Chapter 5).

Placement is a first-order control (measured). The same absorber has 7.34 times more effect moved up than moved down, while changing its spectrum at a fixed placement is worth a factor of 4.95, and spectral shape stops mattering once the absorber leaves the responding layer (Experiments 7, 8 and 10).

The chemistry, not the modeller, decides placement, and the closest gas-phase candidate fails on it (measured). At the profile its own chemistry gives it, ClS_2 leaves the response band and the radiation field becomes that of no absorber, to 0.001487% (`id.l2_ratio_pct`; Experiment 9).

No species the model can represent passes all three tests (calculated from saved model output). The screen uses only saved model output and stored cross sections (Experiment 12).

Two published candidates are reassessed (calculated from published data). The proposed iron and sulfur minerals cannot absorb enough at 375 nm even as pure solids, and the published rejection of ferric chloride does not survive if ferric chloride absorbs in the droplets as it does in hydrochloric acid (Experiments 13 and 14). Neither result uses the model, and neither identifies the absorber.

The chemical responses are small and mostly unresolved (bounded). Apart

from SO and SO_3 when the absorber is moved up, by 8.404% and 4.718% (keys `res.e7a.SO.inband` and `res.e7a.SO3.inband`), the changes in the sulfur reservoirs inside the band are at or below the chemistry floor, or above it by too little to settle anything (SO in Experiments 3 and 4B), and are reported only as upper bounds.

7.2 Limits

1. **One column, one sun angle.** The model is a single global average at a zenith angle of 58 degrees, with every photolysis rate halved for day and night (Table 3.1). Latitude and local time are not represented, and every run is a steady state.

2. **Low vertical mixing.** Every run uses the Bierson and Zhang mixing profile [2], orders of magnitude below what has since been derived for the clouds from observations and models [3, 11]. The vertical wind is switched off. The placement verdict of Experiment 9 has therefore been tested at one mixing strength only; Chapter 9 proposes to test it at others.

3. **Simplified cloud droplets.** The cloud droplets enter the radiation only through fixed, fitted cross sections for two droplet modes; the full droplet calculation in the code stays switched off in every run (Section 3.5).

4. **An empirical reference.** The empirical absorber's spectrum and profile were fitted to the observed brightness of Venus. Spectral rankings measure resemblance to that fit, not correctness (Experiments 8 and 11).

5. **Only what the network contains.** The screen can judge only species the model tracks and holds cross sections for: 23 of 62. Ferric chloride and the iron minerals have no chemistry in the model and could not be placed by it.

6. **A floor from one pair.** The chemistry floor comes from a single pair of replicate runs and is a lower bound on the true numerical scatter (Section 4.6).

None of these limits weakens a comparison between two runs, which share them; they limit how far the verdicts carry to the real atmosphere, and the first two are the subject of Phase 2.

Chapter 8

Verification

No number in this report was typed in by hand. This chapter shows how numbers are produced and checked, how to read a registry key, and what the checks catch.

8.1 The registry and its keys

The script `verify/recompute_25m0328.py` reads the model output committed to the repository, recomputes every quantity the project uses, and writes them to `verify/registry.json`: 10830 named values in 71 groups. Anyone can rebuild it from the committed output by running the script, which also states the conventions of Chapter 4.

Each value has a *key*: parts separated by full stops, read from left to right, from the general to the particular, *group* . *case or species* . *quantity* . *qualifier*. The group names the source, usually an experiment; the middle parts narrow it to one run, species, reaction or window; the last part names the quantity and, where needed, a wavelength or a unit. Not every key has all four parts. Table 8.1 lists the groups quoted in this report, and Appendix A lists, for each experiment, its run files, scripts and registry groups.

Example 1. `e12.S2O2.short.360`: Experiment 12; species S_2O_2 (OSSO); shortfall factor; at 360 nm. Its value, 2.4563×10^5: the model holds that many times too little OSSO above 69 km.

Example 2. `res.e7a.SO.inband`: reservoir responses; case 7A, the absorber moved up; species SO; largest relative change inside the response band. Its value, 8.404%, is one of the two abundance changes above the chemistry floor.

Example 3. `e7.7a.SO2.R781.col` and its twin ending `col_km`: Experiment 7; case 7A; species SO_2; reaction R781; the column-integrated change in its rate, with thickness in centimetres, 8.2352×10^8, and in the old, wrong, kilometre form, 8235.1671, kept to show the error (Section 4.2.4).

Table 8.1: Registry groups quoted in this report.

Group	What it holds
`e1` to `e14`	One experiment each; letters mark a case or sub-study (`e7a` is Experiment 7A, `e3fp` the whole-field test of Experiment 3 at all four amplitudes)
`res`	Largest change in a sulfur reservoir species, per case
`dJ`, `dJy`; `dJz`, `dJyz`	Photolysis change as ΔJ and as $\Delta(Jn)$; the heights where each peaks
`nom`, `conv`	Reference and baseline runs; steady-state tests
`pnom`, `p9`, `spec_*`, `tau`	Absorber profiles, spectra, and optical depths built from them
`scr`, `iso`, `corr`	Experiment 8 screen, Experiment 11 isomers, spectral correlations
`five`, `share320400`	Comparison of the five spectra; share of absorption between 320 and 400 nm
`falselead`	The optical depths of the false lead (Chapter 3)
`id`, `chr`, `chr8`, `twoby2`; `const`, `p2`	Tests of Experiments 9 and 10; fixed constants; Phase 2 mixing times

Example 4. `e1.aflux.meanrel`: Experiment 1; actinic flux; mean relative change. It is stored as the fraction 1.3553×10^{-4} and printed as 0.013553%. Keys ending in `pct` hold percentages; keys ending in `rel` hold fractions.

8.2 The checker, and what it cannot catch

The script `verify/check_documents_25m0328.py` finds every number in a document written with three or more decimals, in scientific notation, or as a percentage with two or more decimals, and accepts it only if a registry value, rounded to the number's own significant digits, gives it (Section 4.1). Three faults in the checker (unbraced exponents, too tight a tolerance, DOIs read as numbers) were found because it rejected correct numbers, and fixed. Chapters 1 to 10 of this report print 407 checked numbers and the twenty figures print 129 (checked by `figures/check_figures.py`); none is unmatched.

The checker proves that a number is in the registry, not that it came from the right key. For example, `pnom.col_dzw` holds 4.0775×10^{8} per square centimetre, which the checker would accept as the absorber column; but only `pnom.column`, 4.0776×10^{8}, summed on the grid the absorber term uses, reproduces τ_{req}. Every number in this report was therefore also checked by hand against the key named beside it.

Reproduction. The eleven runs in the `conv` group, from the nominal run of Experiment 2 to Experiment 11, all ended with `longdy` between 0.0097778 and 0.0099988, below the limit of 0.01. The registry can be rebuilt by anyone from the committed output with `verify/recompute_25m0328.py`.

8.3 One number, traced from end to end

Take the 7.34 of Experiment 7, the number this report quotes most often, and follow it from the model to the page.

1. **The model output.** Three runs saved in the repository's `output` folder: the upward move, `Experiment7A_UUV-upward18-25m0328.vul`; the downward move, `Experiment7B_UUV-downward18-25m0328.vul`; and the reference, `Experiment2_A1_UUV-nominal-25m0328.vul`, with the absorber in its usual place. Each file holds the actinic flux field of its run.

2. **The calculation.** `verify/recompute_25m0328.py` loads the three files and calls its function `radiative()` twice, once for each move against the reference. The function subtracts the two fields and evaluates Equation (4.5).

3. **The registry.** The results are stored as `e7a.L2pct` and `e7b.L2pct`, and their ratio, 7.3367, as `e7.ratio_7A_over_7B`.

4. **The sentence.** Section 5.6 prints 2.2498% and 0.3067% with those keys, and the ratio as 7.34.

5. **The checker.** It finds 2.2498% and 0.3067% in the chapter and accepts each, because the registry value rounded to the same digits gives it. The ratio, printed with two decimals, falls outside the checker's rule, so it was checked by hand: 2.2498 divided by 0.30665 is 7.3367, which rounds to 7.34.

Any number in this report can be followed the same way, backwards from the key printed beside it.

8.4 What the checks catch

Recomputing every number and checking every printed value guards against three kinds of error, and each can be seen in the registry or the code. **Units**: an integral over altitude taken with the thickness in kilometres is 10^5 times too small; the registry keeps both forms of one such integral, `e7.7a.SO2.R781.col` and `e7.7a.SO2.R781.col_km`, so the size of the error can be seen (Section 4.2.4). **The wrong key**: a number can be in the registry and still answer a different question, which is why every sentence is also checked against the key it names (Section 8.2). **Reading below the floor**: a change in abundance smaller than 2.5278% is not a measurement, so it is reported only as an upper bound (Section 4.6).

Chapter 9

Phase 2 Proposal

Phase 1 found that placement decides whether a candidate matters, but every run used one weak description of vertical mixing (Chapter 7). Phase 2 asks whether the verdict survives realistic transport and whether transport can carry a candidate into the response band. It also takes up the detailed modelling of ferric chloride, which the supervisor reserved for this phase.

9.1 The idea: fast winds, slow rising

The supervisor's direction, paraphrased: Venus rotates slowly, so its clouds are carried round quickly while air rises slowly; if something from below reaches the cloud top and absorbs ultraviolet light there, it should be looked for. A rough time for mixing to carry a gas across a layer of thickness L is L^2/K_{zz}, with K_{zz} averaged over the layer.

Figure 9.1 shows the result. With the mixing profile of every Phase 1 run, a gas needs about 552 days to cross the layer from 55 to 69 km (`p2.gap.Bkzz.t_days`) and 874 days to cross the response band. The cloud layer goes round the planet in 4 to 6 days [12], and one Venus solar day lasts 117 Earth days [1]. While mixing carries a parcel across the 55 to 69 km gap, the winds take it round the planet 92 to 138 times, through 4.7 solar days. These times are rough, but the contrast does not depend on the details.

9.2 What the literature adds

The absorber is brought up, not made at the surface. Bertaux et al. found that the cloud-top wind and ultraviolet brightness at low southern latitudes vary with the mountains of Aphrodite Terra below. Wind over the slopes launches gravity waves that break just below the cloud top; where the wind speeds up, air must rise to make up for it, and that upwelling brings the absorber to the cloud top. They also note that SO_2 falls from 150 ppmv at 54 km to 0.1 to 1 ppmv at the cloud top, so

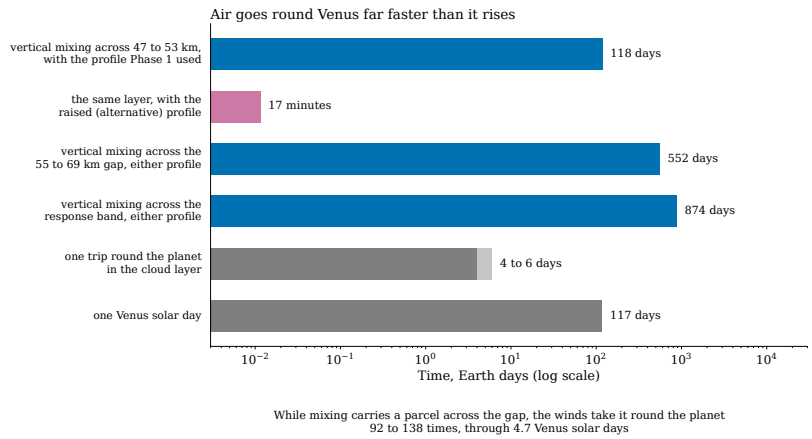


Figure 9.1: Air goes round Venus far faster than it rises. Mixing times L^2/K_{zz} across three layers, from the model's own mixing profiles, against the time for one trip round the planet [12] and one Venus solar day [1]. Keys `p2.X.Bkzz.t_days`, where X is the layer (`raised`, `gap` or `band`), and `p2.raised.BDkzz.t_minutes`.

if the absorber comes from SO_2, it "must be produced inside the upper cloud, just below the cloud top level" [1]. The surface launches the waves; it does not supply the molecule. Titov et al. tie the dark low-latitude markings to the same supply from below [16].

Mixing in the clouds is far stronger than Phase 1 assumed. From Venus Express sulfuric acid vapour, Dai et al. derived a global mean K_{zz} peaking at 5×10^8 cm^2 s^{-1} near 48 km and falling rapidly above 54 km [3]. Lefèvre et al. found 10^6 to 10^8 in the 48 to 55 km convective layer [11].

A convective layer may sit inside the response band at noon. In the model of Lefèvre et al., at the equator at noon, solar heating by the absorber itself drives a convective layer at about 66 to 74 km, which the authors call still speculative [10], overlapping the response band (Figure 9.2). There, a species whose chemical lifetime exceeds the 6000 s convective time becomes well mixed [10]. The authors note that the absorber's uncertain distribution "could have a strong effect on the cloud-top convection depth and vertical mixing", and call for "new 1D chemical simulations with a more realistic convective mixing" [10]. Phase 1 studied exactly that distribution and its radiative effect.

9.3 What the model already contains

Three switches in the supplied code bear directly on Phase 2.

A second mixing profile, never used. The configuration file uses the mixing in `atm/atm_Venus_Bkzz.txt`. A second file, named in a line that is commented

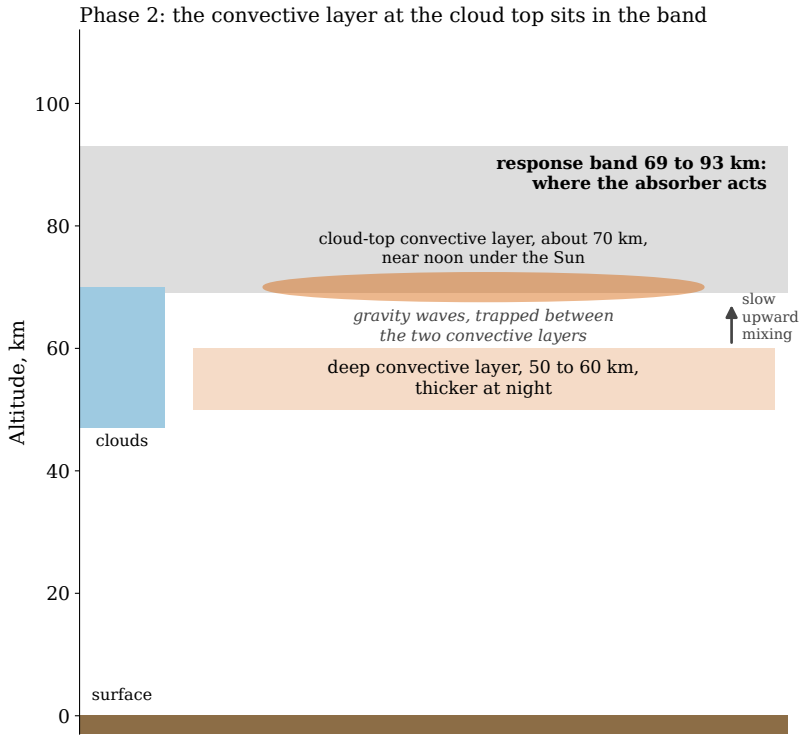


Figure 9.2: Where the layers sit. The deep convective layer, 50 to 60 km in this study (other set-ups place it at 47 to 56 or 48 to 55 km, as above), is thicker at night; the cloud-top convective layer, near 70 km, appears at noon and lies inside the response band [10]. Keys `e6.interval.90.lo` and `e6.interval.90.hi`.

out, sits beside it: `atm/atm_Venus_B+Dkzz.txt`. The two are identical except at the four layers centred at 47 to 53 km, where the second is raised by 8716 to 14011 times (`p2.kzz.raised.ratio_min`, `.ratio_max`) to the values of Dai et al. [3] (Figure 9.3). With it, the crossing time of that layer falls from 118 days to 17 minutes. Every Phase 1 result was reached without this enhancement. It stops at 53 km, so on its own it does not reach the band.

Vertical wind, coded but off. `use_vz` is `False`, but upwind advection is coded in `op.py`, and `build_atm.py` can read a velocity per level from a file. **Day and night as two numbers.** Photolysis is halved (`f_diurnal`) at one zenith angle of 58 degrees (`sl_angle`), untested for a noon-only layer.

9.4 Work packages

Each work package below is set out the same way: what changes in the model, what is measured, what the result would mean, and what it needs first. Every measurement uses a quantity Phase 1 already computes, so the registry and checker carry over unchanged.

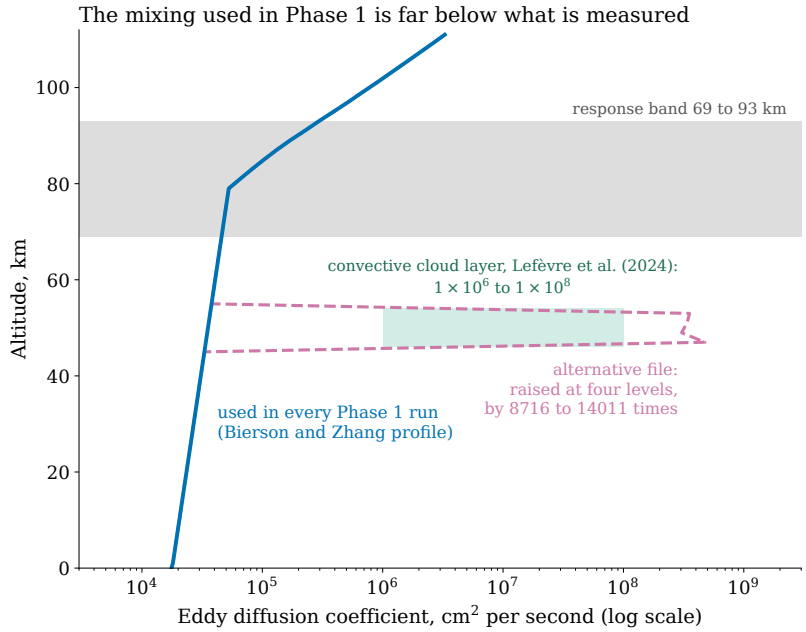


Figure 9.3: The two mixing profiles supplied with the model. Solid: the profile of every Phase 1 run [2]. Dashed: the unused file, raised at four layers. Shaded: the convective layer of Lefèvre et al. [11]. Keys `p2.kzz.raised.ratio_min` and `p2.kzz.raised.ratio_max`.

WP1. Does the Phase 1 verdict survive stronger mixing?

- *Changes:* the mixing file named in `vulcan_cfg.py`, switched to the raised profile of Section 9.3; then a sweep of K_{zz} from 10^4 to 10^8 cm^2 s^{-1} over 46 to 70 km.
- *Measures:* each candidate's centroid, its share of the column in the response band, and ΔF (Equation (4.5)), with the existing registry script.
- *Would mean:* if the centroids and shares hold, the placement verdict survives the largest uncertainty in transport; if a candidate rises into the band, it goes back to the amount test. Dai et al. found the raised profile's effect on abundances "slight" [4]; the new question is placement and radiation.
- *Needs:* nothing new; the file and the code exist.

WP2. A cloud-top convective layer in one dimension.

- *Changes:* a new K_{zz} profile that adds the noon layer near 70 km at the strength reported by Lefèvre et al. [10, 11], justified against their published values.
- *Measures:* the quantities of WP1, and the placement screen of Experiment 12 rerun on the new output.
- *Would mean:* whether mixing inside the band can hold a candidate there. This is

the piece that reaches into the band.

- *Needs:* WP1, so that the effect of the new layer is separated from that of stronger mixing lower down.

WP3. Upwelling.

- *Changes:* `use_vz` switched on, with a vertical velocity for each level read from a file by `build_atm.py` and used by the upwind advection already coded in `op.py`.
- *Measures:* the inverse question: the smallest upward speed over 55 to 67 km that delivers each candidate into the band before its chemistry destroys it.
- *Would mean:* compared with what wave-driven upwelling could supply [1], whether rising air can do what mixing cannot. A one-dimensional model cannot represent a wave breaking over a mountain, so this tests a necessary condition only.
- *Needs:* the lifetimes of WP5; it runs alongside WP2.

WP4. The subsolar column.

- *Changes:* `sl_angle` and `f_diurnal` set for a column under the noon Sun instead of the halved, averaged sunlight, with a second column without sunlight for midnight, both on the mixing of WP2.
- *Measures:* the quantities of WP1 at noon against the global mean, and SO_2 at noon against the observed noon decrease of SO_2 that accompanies an increase of the absorber [10].
- *Would mean:* whether a candidate that fails in the global mean passes where light, heating and mixing are strongest together.
- *Needs:* WP2.

WP5. A fourth test: lifetime against transport.

- *Changes:* nothing in the model; a read-only script over saved output.
- *Measures:* each candidate's chemical lifetime at each height, from its loss terms in `chem_funs.py` read as in Experiment 5, against the time to reach the band, L^2/K_{zz} (Section 9.1), with the 6000 s convective time at 70 km [10] as a mark.

- *Would mean:* a fourth axis for the screen of Experiment 12, and the answer to why the chemistry keeps ClS_2 low (Experiment 9).
- *Needs:* the output of WP1.

WP6. Ferric chloride.

- *Changes:* iron and chlorine chemistry added to the network (a meteoric iron source, reactions with hydrogen chloride, uptake into droplets), and the ferric chloride cross sections obtained from the laboratory group.
- *Measures:* the placement test of Experiment 9 for ferric chloride: its centroid, its share in the band and ΔF.
- *Would mean:* whether ferric chloride sits where the absorber acts, the test that Experiment 14 could not make.
- *Needs:* the transport framework of WP1 to WP4; some rate constants may not exist, which is why it comes last.

Order and risks. WP1 and WP5 come first and can give a publishable result on their own; WP2 and then WP4 follow, WP3 alongside WP2, and WP6 last. Three risks are stated now. The absorber's heating drives the mixing that would move it, so the test will be run with the mixing that heating would give and without it, and both reported. The ferric chloride preprint's group is active [8], so WP6 may be overtaken. And the 55 to 69 km gap may prove uncrossable; if so, that is the result.

Chapter 10

Conclusion

10.1 Key Findings

The model's absorber had been switched off, and switching it on locates its effect. The absorber term of VULCAN-Venus was present but inactive. Once active, it changes the radiation field in proportion to its amount, and the change lives in one band of altitude, 69 to 93 km.

Where an absorber sits matters more than its spectrum. The same absorber has 7.34 times more effect moved up than moved down, against a factor of 4.95 for its spectrum at a fixed height.

The chemistry decides where a candidate sits. Placed by the model's own chemistry, ClS_2, the gas closest in amount, falls below the response band and becomes radiatively the same as no absorber. Being made in the right amount and absorbing at the right wavelengths is not enough; a candidate must also be in the right place.

No species in the model passes all three tests. Of 23 species that could be screened, none supplies enough absorption, and the three tests disagree about which comes closest.

Two published candidates are reassessed. The proposed iron and sulfur minerals cannot absorb enough at 375 nm even as pure solids. The published rejection of ferric chloride rests on a measurement in water; if ferric chloride absorbs in the droplets as in hydrochloric acid, it is a plausible minor component, though its altitude is untested.

Every number is verified. The registry recomputes every quantity from the saved output, the checker tests every printed number precise enough to test, and Chapter 8 traces numbers to the code.

Phase 2 asks whether these verdicts survive the much stronger mixing now estimated for the Venus clouds, and whether it can carry a candidate into the band.

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Appendix A

File Map

This appendix lists, for each experiment, the files in the repository that hold its runs and inputs, and the registry groups that hold its numbers. Every run file is in `output/`, and the full settings of each run are recorded beside it in `output/cfg_<run>.txt`. Every number is recomputed from these files by `verify/recompute_25m0328.py` into `verify/registry.json` (Chapter 8). Run names below omit the common ending `-25m0328.vul`.

Table A.1: Where each experiment lives in the repository.

Experiment	Runs (in `output/`)	Other inputs and scripts	Registry groups
Baseline	`Nominal_Bkzz_SO2-25m0328-reference`, `-baseline`, `-laptop-baseline`	`atm/UV_absorber.txt`, `atm/mode1+2.txt`	`nom`, `pnom`, `falselead`
1	`Experiment1_A0_UUV-control`		`e1`
2	`Experiment2_A1_UUV-nominal`		`e2`
3	`Experiment3_A0p5_UUV-half`, `Experiment3_A2_UUV-double`	`analyze_Experiment3_sensitivity_final_25m0328.py`	`e3`, `e3a`, `e3b`, `e3fp`
4A	runs of Experiments 1 and 2	`Experiment4A_exact_spectral_attribution_final_25m0328.txt`	`e4a`
4B	`Experiment4B_B1_300-320` to `Experiment4B_B6_420-698`	`atm/UV_absorber_Experiment4B_B<n>_25m0328.txt`; `output/Experiment4B_consolidated_analysis_25m0328.txt`	`e4b`
5	runs of Experiments 1 and 2		`e5`
6	runs of Experiments 1, 2 and 4B		`e6`, `e6ctl`
7	`Experiment7A_UUV-upward18`, `Experiment7B_UUV-downward18`	`atm/mode1+2_Experiment7A_shift18_25m0328.txt`, `atm/mode1+2_Experiment7B_shiftDown18_25m0328.txt`	`e7`, `e7a`, `e7b`, `p7a`, `p7b`, `specband`
8	`Experiment8_baseline_A1`, `Experiment8A_ClS2`, `Experiment8B_OSSO_S2O2`, `Experiment8C_S3`	`atm/UV_absorber_Experiment8<A, B, C>_*.txt`; folder `Experiment8-9/`	`e8`, `e8a` to `e8c`, `scr`, `corr`, `spec_8*`
9	`Experiment9_ClS2-selfconsistent`	`atm/mode1+2_Experiment9_ClS2profile_25m0328.txt`; folder `Experiment8-9/`	`e9`, `p9`, `id`, `dJ`, `dJy`, `res`
10	`Experiment10_S3-on-ClS2profile`	folder `Experiment10/`	`e10`, `chr`, `chr8`, `twoby2`
11	`Experiment11A_SSCl2`, `Experiment11B_ClSSCl`	`atm/UV_absorber_Experiment11<A, B>_*.txt`; folder `Experiment11/` with its `data/`	`e11a`, `e11b`, `iso`, `five`, `share320400`
12	none; reads the saved baseline	folder `Experiment12/`	`e12`
13	none; published data	folder `Experiment13/` with its `data/`	`e13`
14	none; published data	folder `Experiment14/` with its `data/`	`e14`
Phase 2	none	`atm/atm_Venus_Bkzz.txt`, `atm/atm_Venus_B+Dkzz.txt`	`p2`